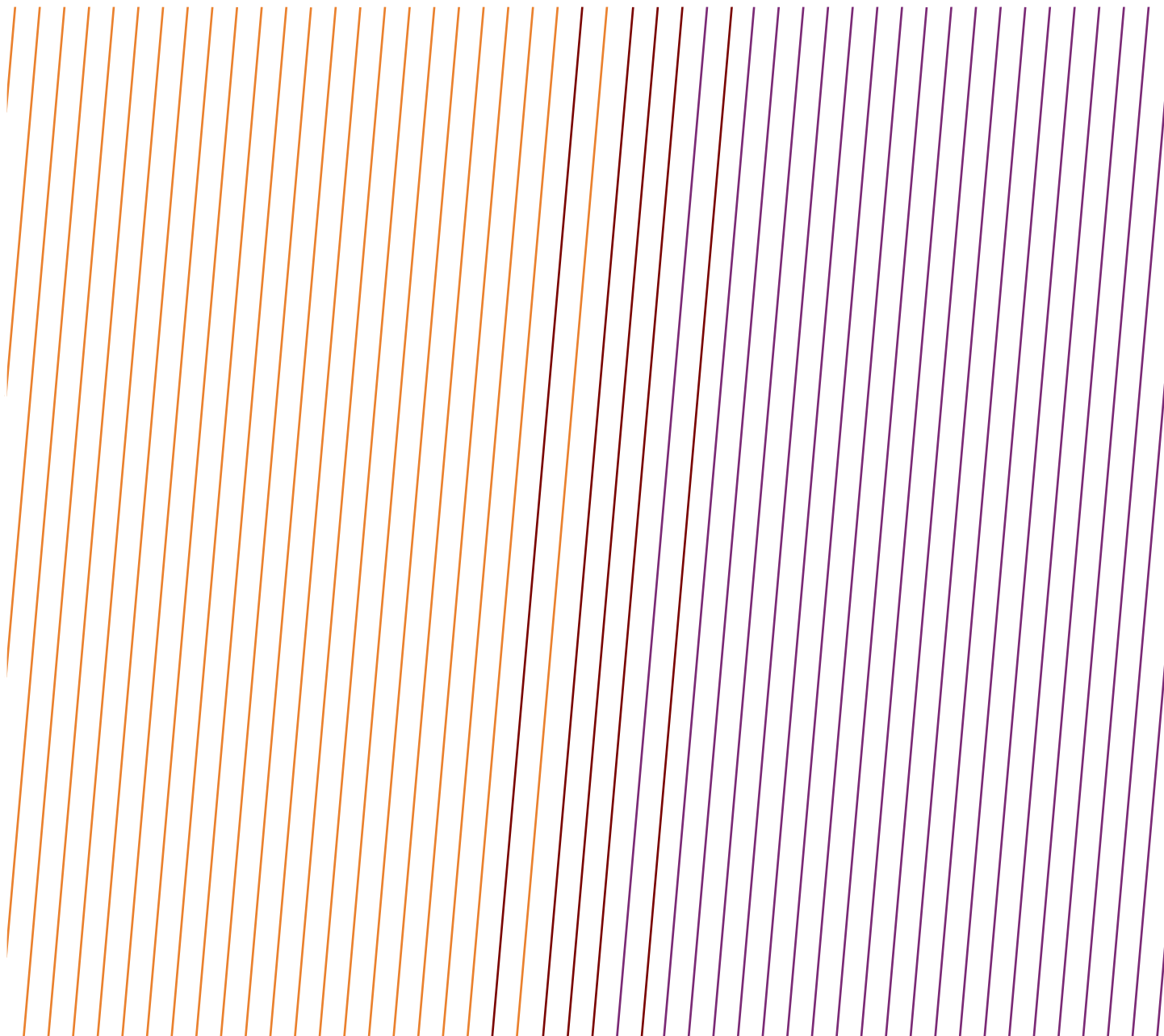


DATA SERIES

Safety performance indicators – Process safety events – 2020 data

Tier 1 PSE, fatal incident and high potential event reports

*Tier 1 PSE, Fatal incidents and high potential events that were also process safety events (PSE),
and fatal incidents and high potential events that were PSE-related – 2020*



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DATA SERIES

Safety performance indicators – Process safety events – 2020 data

Tier 1 PSE, fatal incident and high potential event reports

Tier 1 PSE, Fatal incidents and high potential events that were also process safety events (PSE), and fatal incidents and high potential events that were PSE-related – 2020

Revision history

VERSION	DATE	AMENDMENTS
1.0	January 2022	First release

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TIER 1 PROCESS SAFETY EVENTS

Tier 1 PSE are predominantly lagging indicators related to Loss of Primary Containment (LOPC) referred to as a Process Safety Event (PSE). The Tier 1 KPI records events with greater consequence within the four-tier approach. For more information see the introduction and IOGP Report 456, *Process Safety – Recommended Practice on Key Performance Indicators*.

IOGP has been gathering Tier 1 PSE narrative reports from its Members since 2013. These include Tier 1 PSE reported both by companies and contractors. Event reports are categorised by region, country, location (onshore/offshore), cause, activity at the time of the event, and, from 2019 onwards, point of release.

The information provided here is not detailed; often the root cause of an incident cannot be established. However the information should assist organisations to identify likely hazards, human factor issues and failure modes that they may not have recognized within their own operations. In particular it allows organisations to question whether their own Safety Management System would have prevented the event occurring and mitigated its consequences.

The database from 2020 onwards is now available and searchable at <https://data.iogp.org/ProcessSafety/Tier1PSE>.

Note that a descriptive report has not been provided for every Tier 1 PSE reported.

This database is a tool for learning and should not be considered a complete record of Tier 1 PSE in the upstream oil industry or the IOGP Membership.

TIER 1 PROCESS SAFETY EVENTS

AFRICA ONSHORE

DATE: May 13 2020

COUNTRY: Congo

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During a tour in the tank area, the operator observed a rise of crude oil through the ground. The transfer between the two tanks was stopped. The volume of crude oil + water released was 135 cubic metres in which the volume of oil was 23 cubic metres.

WHAT WENT WRONG?:

Hole in the line between the two tanks.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Nov 29 2020

COUNTRY: Congo

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Loss of containment was reported following a breakthrough of an isolated line.

WHAT WENT WRONG?:

Corroded infrastructure.

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Oct 15 2020

COUNTRY: Egypt

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A high potential event happened while retrieving the probe of the analyser through the DBB valve installed on the sweet gas line for trouble shooting by Instrument team under a PTW, the probe was ejected with high pressure "75 bars" to the air close to the worker. This resulted in a release of sweet gas to the atmosphere with high pressure (LOPC). Immediately the area was barricaded, the Production was shifted to the Metering station, a new valve was installed after depressurization and situation was controlled. No injuries.

WHAT WENT WRONG?:

- Lack of knowledge how "overall system" (e.g. instrument, DBB, high-pressure pipe) works from all involved parties. This indicates a failure of the management systems controlling the work.
- The PTW that was issued to control this work did cover the normal removal of the probe; however, the attached work instructions and risk assessments did not adequately address all the abnormal potential hazards.
- The concerns raised by field workers to their supervision, including all disciplines, failed to be recognized as a serious hazard.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N.A.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Execution of activities

TIER 1 PROCESS SAFETY EVENTS

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: Aug 11 2020

COUNTRY: Egypt

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Unspecified - other

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Equipment: Pump

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During the completion of a normal Planned Maintenance (PM) activity on the Sulphuric Acid Injection Pump by the maintenance contractor, under a PTW one of the workers accidentally stood, bumped, knocked a small bore (1") outlet pipework from the Sulphuric Acid Storage Tank causing the line to separate from the tank body. An amount of high concentration Sulphuric Acid (approximately 96%) was released into the bund, where it was contained. The tank was approximately 38% full and the release was approximately 2.38m³ there was no injury to any of the persons working in the area and the bund was filled with sand to help contain the liquid and to absorb the acid so that it could be removed for disposal. The use of sand was necessary because the neutralization pit cover is higher than the base of the bund and therefore not all the spilled liquid was directed into the pit.

WHAT WENT WRONG?:

1. The pipe is in the wrong place, it is at the perfect height, in the middle of an access route to be accidentally damaged whenever maintenance is carried out on the injection pumps. The design is not good and the installation regarding supports etc. is poor.
2. The repair carried out previously, where changes were made to the material and glue, was not subjected to a Management of Change (MOC) process.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N.A.

TIER 1 PROCESS SAFETY EVENTS

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Congestion, clutter or restricted motion

DATE: Oct 8 2020

COUNTRY: Gabon

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Emergency shutdown

POINT OF RELEASE: Equipment: Pig launcher/receiver

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Following shutdown of export pumps, operator was conducting rounds and noticed oil coming out of the pig receiver door latch, which was still closed. The crude spilled inside the bund wall and within the terminal boundary.

WHAT WENT WRONG?:

No preventive maintenance conducted on pig door and seals, only corrective repairs.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Implement preventive maintenance routine for 'weak' elements of the pig receiver.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Feb 27 2020

COUNTRY: Gabon

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

CATEGORIES:

PRD release to atmosphere above threshold in any 1 hour period and results in:

- liquid carryover or
- discharge to a potentially unsafe location or
- onsite shelter in place or
- public protective measure (e.g. road closure)

INCIDENT DESCRIPTION:

Loss of containment from IGF tank. During rounds, operator noticed dark patch on the ground. On closer inspection oil was observed oil overflowing from the 10" thief hatch on the top of the tank. The operator stopped water injection pump and oil overflow stopped. Approx. 20m3 of crude was contained the concrete bund wall and around the tank.

WHAT WENT WRONG?:

No existing maintenance plan for the involved equipment.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Undertake a design review of the overall system to ensure asset integrity and alarms, control panel response combined with operator surveillance and inspections.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Dec 2 2020

COUNTRY: Nigeria

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

During condensate evacuation, a fire occurred which was put out by fire personnel who were part of the condensate evacuation crew. Two company personnel had minor heat burns and were taken clinic for medical attention. The Condensate evacuation exercise due to a massive condensate influx into the NGC manifold and consequently into the customers' delivery lines was suspended and investigation commenced.

WHAT WENT WRONG?:

Overfill of condensate during condensate evacuation exercise

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Follow procedures and sign off after each step

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate communication

TIER 1 PROCESS SAFETY EVENTS

DATE: Mar 27 2020

COUNTRY: Nigeria

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Platform/Well Pad Flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

An oil leak was observed at the wellhead. The team immediately mobilized spill response materials to the well location. The well was closed in, fully isolated and depressurized at 0915a.m. The spill was contained within well slot right of way and oil recovery process initiated. From initial assessment, the leak source was the Wellhead bean box area. Full investigation into cause of leak ongoing.

WHAT WENT WRONG?:

Rubber particles and sand erosion damaged bean box

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Redefine inspection and repair intervals for bean box

Conduct Risk Assessment.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

TIER 1 PROCESS SAFETY EVENTS

DATE: Mar 29 2020

COUNTRY: Nigeria

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Platform/Well Pad Flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Leak was observed at well site near sentry post. The operations team mobilized to the wellhead, confirmed the leak and closed in the well. The wellhead area was boomed around to contain the spill.

WHAT WENT WRONG?:

Bean Box damage due to erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Redefine inspection and repair intervals for bean box.
- Conduct Risk Assessment.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate supervision

TIER 1 PROCESS SAFETY EVENTS

DATE: Dec 13 2020

COUNTRY: Nigeria

FUNCTION: Production

NUMBER DEATHS: 1

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (not at intended discharge location)

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

Information not available

WHAT WENT WRONG?:

Information not available

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Information not available

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

NONE: Unspecified

DATE: Jun 16 2020

COUNTRY: Nigeria

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

Fire from the floating roof of crude oil tank (n°3) during heavy rain with thunder and lightning.

WHAT WENT WRONG?:

Failure of the lightning protection system (LPS) due to the lack of maintenance program to ensure functional reliability and sustained availability during lightning strikes.

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Always ensure all lightning protection systems are in place and adequately maintained.
2. Ensure all firefighting equipment are in serviceable condition and regularly tested.
3. In the event of any dysfunction of firefighting equipment when immediate repair is not possible a backup plan that is tested and proven to be reliable should be put in place.
4. The need to develop a survey programme and periodically conduct a survey and functionality test on all Early Streamer Emission System (ESE) units to ascertain their status and availability on demand.
5. High level competence/skills of personnel involved during emergency is fundamental in managing emergencies

BARRIERS:

Hardware Barrier Failures: Ignition Control

Hardware Barrier Failures: Protection Systems - including deluge and fire water systems

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Nov 8 2020

COUNTRY: Nigeria

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Spill over from dehydration unit due to delay in switching on oil transfer pump during manual operation.

WHAT WENT WRONG?:

Delay in starting the oil transfer pump during manual operation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N.A.

BARRIERS:

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate supervision

TIER 1 PROCESS SAFETY EVENTS

DATE: Jun 12 2020

COUNTRY: Tunisia

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

On Friday June 12, 2020, during sand removal activities for an escape route commonly understood as a “patrol road” around the wellsite, the underground 6” gas flow line (FL) was punctured by a front loaders teeth as it was levelling the ground. At the time of the incident, the pressure of the gas 6” FL and the methanol 2” FL were respectively approximately 54 barg and approximately 285 barg respectively.

A vapor cloud was immediately generated without ignition. Luckily, no Unconfined Vapor Cloud Explosion (UVCE) or jet fire occurred. The involved contractor team comprising of five (5) persons evacuated the site and moved to a safe location. No injuries occurred.

WHAT WENT WRONG?:

- Unidentified (unmarked) underground services in immediate proximity to sand removal operations.
- Absence of warning tape on top of buried services. The investigation revealed few non-conformances in the construction and gaps vs the as-built drawings.
- Missing key marker pole and absence of warning tapes on top of the FL or Fibre Optic Cable (FOC).
- Widening the escape route to more than required allowed for repetitive access by vehicles over time and the perception that this was a “NORMAL” or “ROUTINE” road that can be subject to normal grading by heavy equipment.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Ensure better Knowledge management from “EXECUTE” to “OPERATE” and prevent knowhow losses: Operation team should ensure more synergy and consultation with remaining project organization during “OPERATE” phase of the project mainly in future MoC or scope of work approval.
- The Management of construction change via technical queries (TQ's) by EPCC is deemed less efficient than a standard Company MoC with a supporting risk assessment. In this particular change of the trench configuration, Projects should have provided additional warning and signalization equipment to denote the underground services which are, in this case, exceptionally close to the natural ground level, and were at high risk of impact by ground disturbance either natural or manmade.

BARRIERS:

Human Barrier Failures: Authorization of temporary and mobile equipment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate communication

TIER 1 PROCESS SAFETY EVENTS

AFRICA OFFSHORE

DATE: Oct 23 2020

COUNTRY: Angola

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During a walkdown inside the engine room a pinhole leak of diesel coming from top of the air duct was identified. The source of the release was from an oil transfer and sludge system line pipework failure. The failure point was a 7mm hole on an elbow weld. At the time of the incident, the DO/Sludge Overflow tank was being emptied and the line was pressurized.

WHAT WENT WRONG?:

The leak occurred because the weld was subject to accelerated preferential corrosion. The bilge fluids contain high levels of bacteria and are corrosive to carbon steel.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Asset to assess other lines that may be susceptible to corrosion from bilge fluids and replace or repair where necessary. Based on this and subsequent findings, it is recommended that the line be replaced with a new GRP; or, Eliminate/Reduce Where this is not feasible replacement should be at least 'like for like' (Carbon Steel) and the line should be corrocoated to reduce effects of corrosion.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

TIER 1 PROCESS SAFETY EVENTS

DATE: Aug 5 2020

COUNTRY: Angola

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

During routine testing, a diesel engine driving a fire water pump caught on fire. The fire initiated from an LOPC on the diesel supply line coming from two pin holes, one 3mm and a second 1mm. The holes were the result of external corrosion.

WHAT WENT WRONG?:

During building of the vessel, the diesel engine OEM issued a technical service bulletin to replace all the diesel lines on the engine. The lines were replaced but were not painted. The unpainted lines resulted in an accelerated corrosion rate.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Ensure a process to manage vendor service bulletins and modifications is in place during the execute phase of major projects and that it defines how ownership is handed over to equipment owners within operations.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Policies, standards and objectives

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: May 31 2020

COUNTRY: Angola

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During a diesel oil transfer from PORT diesel oil storage tank to the STBD diesel oil service tank there was a diesel oil spill inside of the engine room. The spilled diesel oil was contained and cleaned up by site response team. There were no injuries or environmental impact.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

There was a diesel overflow because service tank level instrument failed to correctly indicate the tank level.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review maintenance frequency and priority of all tank level instruments and perform required changes to account for the single instrumented barrier provided by these instruments and to ensure calibration is performed at a sufficient frequency. Consider the addition of an automated high-level trip functionality

BARRIERS:

Hardware Barrier Failures: Detection Systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Mar 8 2020

COUNTRY: Angola

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

On March 8th 2020, at 12:43 hours, gas detectors activated on starboard side near drain tank dump valve and alarms initiated a facility ESD / muster and a system blow-down. Emergency Response Team (ERT) reportedly found 10" dump valve on drain tank open approximately 5mm with corrosion around seat. ERT attempted to fully close valve but unable to establish a seal. Initiated manual venting of blanket gas to reduce tank pressure; re-established water seal in u-tube below dump valve. Analysis indicates that an estimated 3700Kg of Gas was released (undetermined mix of inert and HC) over a 77min period from Drain Tank, until pressure reduced and drain seal re-established.

WHAT WENT WRONG?:

- Primary seal through U-tube water trap; anticipate water seal lost and/or overcome by increasing pressure either due to evaporation or integrity of u-tube.
- Crude oil heaters not performing to design given plate cooler/refurbishment and steam system reliability; modelling suggest entrained gas breakout 6x higher than design in tanks consistent with pressure trends.
- Reid Vapor pressure analysis on oil stabilization not carried out on FPSO.
- Found leak rate across SDV downstream of oil coolers during ESD to exceed design criteria.
- Eroded operating and maintenance practices in cargo Offloading Systems.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Review Marine Systems Integrity program to confirm equipment is included in PM.
- Share learning related to energy associated with low pressure high volume systems (e.g.: cargo tanks).

TIER 1 PROCESS SAFETY EVENTS

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: May 17 2020

COUNTRY: Angola

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

- Gas leak observed from vacuum valve on Inert Gas (IG) purge line.
- Personnel remotely opened high velocity vent valves and closed the aft-end IG purge and supply vent mains; gas alarms initiated followed by facility ESD and crew muster.
- Decanting of slop tank was initiated to reduce system pressure.
- Emergency Response Team (ERA) deployed to fully depressurize and inspect valve; ERT successfully established seal to stop leak.

WHAT WENT WRONG?:

- Monthly vacuum valve function test conducted the day prior to incident with inert gas system near 0 psi; valve integrity reportedly confirmed by operator using visual and audible indicators without portable gas monitor. Reported two independent visual checks following function test during reinserting of tanks at 0.7 psi.
- Leak triggered at just over 0.7psi per pressure data.
- Valve designed to seal with gravity unless under vacuum. Investigation reported grease / corrosion on valve lever; no valve internal inspection since start-up. No record of annual inspections performed.
- No local isolation exists on IG purge system to safely perform vacuum valve internal inspections without full inert system shut-down.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Review Marine Systems Integrity program to confirm equipment is included in PM.
- Share learning related to energy associated with low pressure high volume systems (e.g.: cargo tanks).

TIER 1 PROCESS SAFETY EVENTS

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: Oct 11 2020

COUNTRY: Angola

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Unspecified

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During the loading of the new configuration of the Communication Unit, an oil spill was observed.

WHAT WENT WRONG?:

Losses of containment due to the opening of the production well valves that connect the well tubing with the tree caps.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Reconsider the design of the Smart Tool and other equivalent engineering stations (inhibition/removal of the buttons with large impact).
2. Good practice to be consolidated, as long as some buttons with large impact are kept in the Smart Tool.
3. Recommendation to do the configuration distributions during the day shift.
4. Be aware of the need to have a procedure regarding the preparation of Subsea control systems' activities on site.

BARRIERS:

Hardware Barrier Failures: Detection Systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

TIER 1 PROCESS SAFETY EVENTS

DATE: Aug 30 2020

COUNTRY: Angola

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Flexible hose/piping

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude leaking to the sea and damage of 2nd section of offloading hose was confirmed.

Volume of oil estimated following in situ and aerial survey: 25 cubic metres.

WHAT WENT WRONG?:

Defective offloading hose.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

DATE: Oct 22 2020

COUNTRY: Angola

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Flexible hose/piping

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During offloading activities, an oil spill was observed.

WHAT WENT WRONG?:

Rupture of offloading hose.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

TIER 1 PROCESS SAFETY EVENTS

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: May 6 2020

COUNTRY: Egypt

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Subsea: Subsea pipeline/flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

An oil leakage was observed coming from the 18" offshore trunk line of a platform to the onshore production Plant. The pipeline was immediately isolated, depressurized and the production was switched to another pipeline.

WHAT WENT WRONG?:

- H2S had been registered to be present since a few year prior to the event
- lifespan of the line
- lack of an established and effective preventive maintenance program
- lack of corrosion protection program
- lack of corrosion monitoring program
- missing process monitoring devices such as pressure switches in the line to alert operator in case of LOPC and dropped in the pressure of the line

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Replace old corroded line and lay new subsea line
- ensure installation of a suitable mean of corrosion protection
- ensure affective use of corrosion monitoring
- establish an effective maintenance program (preventive)
- establish an effective corrosion inspection program (preventive)
- verify possibility to remotely control pressure / temperature / flow (DCS)

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Asset design and integrity

TIER 1 PROCESS SAFETY EVENTS

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Dec 3 2020

COUNTRY: Nigeria

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During a planned turnaround, an oil metallic sheen was observed around the platform.

The sheen was approx. 20m width over 500m length with an estimated spill volume of 159 L (1 bbl) for a duration of 35min.

WHAT WENT WRONG?:

1. The gasket was not fully tightened
2. During the tightening and torquing of the flange bolts, the tension of the jack bolt was not fully released
3. The performing authority declared to Production team that the de-spading plan was completed
4. The CCR saw inconsistent LTs values for L-5 bilge wells and did not request to external operators to check in situ.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Define an action plan to reduce the number of pending curative maintenance actions

BARRIERS:

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Poor leadership/Organizational culture

TIER 1 PROCESS SAFETY EVENTS

ASIA / AUSTRALASIA ONSHORE

DATE: Jul 20 2020

COUNTRY: Australia

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

An operator attended a well site on routine rounds. On approach, the operator noticed the HPU (Hydraulic Power Unit) making abnormal noises and began troubleshooting the source of the noise, during which operator's Personal Gas Monitor (PGM) alarmed on high Carbon Monoxide (CO). The operator continued troubleshooting to identify and document the source of the leak as the engine exhaust flange. On return to his vehicle, the operator felt increasingly unwell, and initiated emergency response. The operator was later hospitalized for exhaust inhalation.

WHAT WENT WRONG?:

The engine exhaust flange was loose (chattering) at the 'donut' gasket (spring loaded flange) between the engine manifold and exhaust pipework, causing a release of exhaust gases. The operator continued to troubleshoot the HPU after acknowledging a PGM CO high level alarm. The operator also felt safe to continue the task, where the risk of Carbon Monoxide exposure from the exhaust was perceived as low.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Lessons Learnt:

- Awareness risks posed by exposure to CO and exhaust gases.
- Do workers know PGM alarm limits are for CO?
- Does our relevant documentation (SOP/PET/HAZOP) adequately cover CO and exhaust hazards?
- How are we responding when our gas detectors go into alarm? Does our response change depending on what gas is present (CH₄, CO, O₂, H₂S)? Have we begun to normalize this risk in some areas of our operations?
- Do teams feel empowered to escalate concerns and activate their distress button in an emergency?

Recommendations:

- Ensure CO risk is recognized and understood across the business, including not only HPUs (combustion engines), but also for gas driven and temporary equipment,
- Determine and document flange management requirements for spring loaded joint types,
- Rollout investigation learnings on CO hazards and gas monitoring.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Monitoring, reporting and learning

TIER 1 PROCESS SAFETY EVENTS

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: Feb 8 2020

COUNTRY: Australia

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

The IP had removed his gloves and was in the process of feeling for leaks around a newly installed regulator that the Snoop had failed to find when the regulator catastrophically failed, the bowl and other components of the regulator struck the IP's left hand causing lacerations to the palm and fingers.

WHAT WENT WRONG?:

1. Incorrect regulator selected for installation. Supervisor was aware of the task during the Bales toolbox that immediately follows the joint Pre-start meeting but didn't ensure that workers had planned task including permits and JSA.
2. Failure to follow PTW or Bypass Procedures -Failure to recognize that the N2 systems qualify as a pressurized system that requires a permit under the procedure.
3. No JSA completed to identify hazards and risks or plan task. Work team had failed to understand that the work task had changed and did not assess the hazards and risks. No Front-Line risk assessment such as a Take 5 to act as prompt for hazard identification.
4. Failure to identify MOC process.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. The importance of establishing and compiling with safe systems of work.
2. The importance of supplying workers with adequate training, instruction and supervision.
3. The importance of supplying and verifying the use of fit for purpose equipment.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

TIER 1 PROCESS SAFETY EVENTS

DATE: Aug 24 2020

COUNTRY: China

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Natural gas was leaking from the pressure gauge joint in a valve chamber.

WHAT WENT WRONG?:

Pressure gauge joint problem.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Enhance equipment PM strategy.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

DATE: Oct 15 2020

COUNTRY: Malaysia

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

Prior to vessel loading, ships crew was performing unblocking of wax on the uninsulated section of a loading arm when the wax liquified and stored nitrogen pressure pushed hot liquid wax into a drip tray. The liquid wax splashed back onto the crew member manipulating the steam resulting in injury to lower back calves, buttocks and left hand. The crew was provided First Aid and sent to the hospital for further treatment. The incident resulted in a Lost Work Case (LWC) and is classified a Tier 1 Process Safety Event on the basis of lost workday in line with API 754.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

- When trying to unblock the loading arm into an open area prior to wax loading, the sudden release of blockage of hot wax got splashed onto a ship crew who was applying steam heating at the area.
- Duty Manager of wax loading facility was facing strong push back by ship officer to use standard practice to unblock the Loading Arm by applying steam heating externally with Loading arm connected. Finally Duty Manager conceded to ship officer demand and at the same time over looked the line still with pressure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Improve training and competence of loading master / jetty team.
- Loading master and Jetty team know that zero energy check can be reliably verified and there is a secondary means of verification.
- Build / ensure loading master / jetty team's job competences are directly linked to process safety / process line work and zero energy verification.
- Ensure Loading Master is clear with accountabilities / responsibilities / escalation process.
- All equipment from main header to the loading arm discharge point have preventive maintenance philosophy defined and urgency of prioritization is made clear.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: Jun 2 2020

COUNTRY: Malaysia

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

A fire broke out at a processing unit separating solvent feedstock from synthetic crude. Site First Responders quickly put out the fire using the fire-fighting facilities. No injury was reported and all personnel have been accounted for. The local Fire & Rescue Station was also informed in line with standard notification process.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

- Hot wax above autoignition temperature had been released from external corrosion induced holes at the bottom of a sample cooler coil. The section of corroded coil was exposed to unknown durations of submersion in water due to the design of the drain point. When the wax was released from the process piping it caught fire and because the process was live, the “fuel” continued to flow until the unit was shut down.
- The sample coolers were identified as low risk in the site report, and the potential for external corrosion due to water build up in the cooler coil drum was not recognized.
- Cumulative risk of unblocking the line using steam and the potential for release of process fluids above Auto Ignition Temperature (AIT) were not recognized by the troubleshooting team. The usage of steam to unblock lines had been a practice normalized within the organization and allowed to be undertaken without a defined procedure or temporary Management of Change (MOC) in place.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Design of the sample cooler considered the process fluid and conditions for material selection, however potential for external corrosion leading to failure was not recognized.
- Inspection protocols and frequency for dormant equipment require clear parameters to be measured as well as added rigor of inspection especially before putting the equipment back in service.
- Site analysis needs to be comprehensive enough to properly review “low risk” items and ensure that there aren’t any latent factors that could inadvertently change the risk profile.
- Routine activities that are not closely managed via written procedures may lead to risk normalization and increased exposure.

BARRIERS:

Hardware Barrier Failures: Protection Systems - including deluge and fire water systems

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: Jan 13 2020

COUNTRY: Malaysia

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

On 13 January 2020, a 36" pipeline parted, causing Major LOPC and Major Fire incident located along a slope. The pipeline was found to have parted cleanly seemingly at welded joint and a section of the upstream pipeline was found kinked downhill towards the same direction of the landslide soil movement. Vegetation along the ROW and trees within 200m radius of the fire were burned and destroyed.

TIER 1 PROCESS SAFETY EVENTS

CCR tried to close the upstream Block Valve (BV) remotely but failed. It could only be closed locally a few hours later due to unfavourable weather and poor access due to bad road condition. The fire extinguished by itself after BV were isolated and blowdown completed.

According to investigation findings, the most probable cause of incident has been identified to be external forces acting on the pipeline resulted from global landslide due to the following:

1. Deterioration of existing slope from severe erosion.
2. Disturbance caused by construction activities at the slope toe as preparation for Soil Investigation and repair of the slope.
3. Increased rain intensity during the period leading to the date of incident.
4. Disruption made by road access construction activity on the existing natural drainage system and displacement of temporary timber logs have caused instability on the already deteriorated slope.

WHAT WENT WRONG?:

The underlying causes identified:

1. Inadequate resources and facilities to take up emerging and increasing P1 action item from comprehensive inspection and geo-hazards risk assessment done after a previous incident in 2018.
2. People involved in Job Hazard Assessment (JHA) review and approval were not trained on geohazards.
3. Insufficient manpower to supervise concurrent critical activities.
4. Insufficient review process and approval by qualified personnel on Contractor Document i.e. Method Statement.
5. Inadequate standards/guideline for temporary works including access road.
6. Unavailability of guideline for Pipeline Operations to shut down gas transmission based on geo-hazards threats.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. To ensure adequate assessment on slope failure and drainage - adequacy by conducting technical assessment on pipeline sections exposed to or within global failure plane/area and review adequacy of drainage at surrounding slopes.
2. Manpower Resources and Competency Management - to conduct training on geohazard for all technical personnel in operations and project involved for pipeline operations/maintenance/repair.
3. Enhance Maintenance Management on Critical Equipment.
4. Enhance Operating Procedure and Operational Response.
5. Enhance work procedure and guideline related to construction and road access at slope areas.

BARRIERS:

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

TIER 1 PROCESS SAFETY EVENTS

DATE: Dec 3 2020

COUNTRY: Malaysia

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

On 3rd December 2020, approximately around 0831 hours, a gas pipeline parted. Based a on quick check with the pipeline monitoring system, it was observed that there was unusual pressure drop between Block Valve (BV)106-107 section.

WHAT WENT WRONG?:

Still under investigation

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Still under investigation

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

TIER 1 PROCESS SAFETY EVENTS

ASIA / AUSTRALASIA OFFSHORE

DATE: May 11 2020

COUNTRY: Australia

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Flexible hose/piping

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from loading hose on reel into a banded area below the reel.

WHAT WENT WRONG?:

Unknown failure mode meant no prompt for close visual inspection. OEM were aware of the failure mode, but WEL were not informed.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Better understanding of failure modes of offtake hoses.

A closer relationship between OEM and Company.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Feb 25 2020

COUNTRY: Indonesia

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (not at intended discharge location)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Gas release from failed tubing and ¾" injection valve on the high pressure compressor.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

Failed equipment (tubing and injection valve) had experienced high cycle, relatively low stress vibrations as result of prolonged PSV lifting, leading to fatigue failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

PSV early lifting is very important for closed-loop PSV installations to prevent further escalation of prolonged undetected PSV lifting as it could happen within safe operating limit. The design of associated small-bore piping assemblies (tubing and branch piping) with PSVs shall be taken into account considering the impact of PSV lifting during an MOC review. The integrity of tubing material/connection systems associated with pilot operated-relief valves shall be maintained systematically as part of the equipment strategy.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

DATE: May 6 2020

COUNTRY: Malaysia

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Planned shutdown

POINT OF RELEASE: Equipment: Compressor/blower/fan

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

While maintenance work was being carried out under a valid e-PTW, the mechanical engineer (Injured Person, IP) was removing the Dry Gas Seal (DGS) from a gas compressor. Suddenly DGS was forced out of the stuffing box and hit the right inner thigh of the IP. DGS approximate weight was 40kg. Emergency Response Team was deployed to the site to rescue the IP then medevacked him to the hospital for further treatment.

WHAT WENT WRONG?:

- Isolation not done as per procedure in combination with isolation valve passing, leading to pressure built up in stuffing box
- Emerging work plan for Drive-End DGS after the completion of Non Drive-End DGS replacement. (Inadequate risk review).
- Failed to monitor and verify the integrity of the isolation prior to breaking containment.
- Improper working position. IP was not completely out of the line of fire.

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Ensure knowledge & competency of isolation philosophy are adequate prior executing work. If there's clarity needed, STOP and suspend the work and seek technical support.
- Before breaking containment – always conduct the final check for all possible trapped pressure that may increase the risk of the activity.
- Review work area and body position prior work execution to minimize the risk.
- If there's a change in work scope, re-conduct the safety and integrity verification to ensure all risk have been mitigated.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

TIER 1 PROCESS SAFETY EVENTS

EUROPE ONSHORE

DATE: Sep 11 2020

COUNTRY: Croatia

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During regular site visit, the operator noticed that the flare was extinguished and immediately after an explosion was heard. After on-site inspection it was found that a flame arrestor had exploded. Compressor station was shut down immediately. On-site inspection showed that there was no damage on flare site piping. Pilot Flame operation was maintained in order to eliminate the opportunity of HC gas concentration increase in the working area. New flame arrestor and pressure monitoring with alarm set points implemented before re-start of the compressor station.

WHAT WENT WRONG?:

Equipment reliability.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Improvement of inspection and maintenance.

BARRIERS:

Hardware Barrier Failures: Detection Systems

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate supervision

TIER 1 PROCESS SAFETY EVENTS

DATE: Dec 22 2020

COUNTRY: Hungary

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Based on a public announcement, the disaster management notified the production manager of the area that a tractor had torn off a line during ploughing. After notifying the moving operators, the well was closed. Area was fenced.

WHAT WENT WRONG?:

Error enforcing conditions (external).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Protection Systems - including deluge and fire water systems

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

DATE: Dec 2 2020

COUNTRY: Hungary

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

There was a leak on the gas inlet pipeline of the boiler. The gas cloud found an ignition source and detonated. A jet fire formed between the two boilers. There were no gas detectors in the boiler building. The gas flow was stopped by closing the pipeline at the pressure reductor. The fire was extinguished until the municipal fire fighters arrived. No personal injury occurred.

WHAT WENT WRONG?:

Insufficient design, layout.

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

All boiler houses were checked with special attention on the inlet pipelines and gas detectors.

BARRIERS:

Hardware Barrier Failures: Detection Systems

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: Jan 3 2020

COUNTRY: Hungary

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During the maintenance of the pipeline, there was an oil runoff. The operator found that the non-return valve that closed the pre-water separator system did not keep the liquid and produced it back to the pipeline.

WHAT WENT WRONG?:

Equipment reliability.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Evaluation of all the non-return valves in all the operations.

BARRIERS:

Hardware Barrier Failures: Protection Systems - including deluge and fire water systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

TIER 1 PROCESS SAFETY EVENTS

DATE: Aug 31 2020

COUNTRY: Italy

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

The event involved a contractor who, while the mechanical reassembly of one of the dosing pumps of the regenerating solutions of the demi water production plant (acid dosing pump) were in progress, accidentally came into contact with sulfuric acid present in the section of piping between the flanged coupling in delivery to the pump and the drain line of the same. The worker underwent first aid. The patient was eventually discharged with a temporary disability prognosis of 14 days.

WHAT WENT WRONG?:

- Maintenance can be improved
- Supply and installation of unsuitable material for the specific maintenance activity
- Incorrect assessment of the chemical risk
- Inadequate training
- Lack of safety culture
- Ineffective training on specific risks and emergency management
- Design can be improved

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Check that the material used is compliant / suitable for the specific maintenance activity before its installation
- Whenever there are variations to the working conditions defined during the preparatory phase, the Stop Work Authority (temporary suspension of activities) must be applied. The application of this tool allows the revaluation of risks and the possible definition of additional mitigating measures.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Emergency Response Equipment and Systems

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

TIER 1 PROCESS SAFETY EVENTS

DATE: Oct 26 2020

COUNTRY: Norway

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During reset of booster compressor, the ball valve downstream was opened and it resulted in gas from hp flare being introduced towards booster compressor 2 and there was gas leakage when several bleachers were not closed/reset.

WHAT WENT WRONG?:

Ball valve was left open that allowed HP flare gas to enter opened bleachers and building.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Revise operating and maintenance procedures. Revise operator competency assurance process.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

TIER 1 PROCESS SAFETY EVENTS

DATE: Aug 19 2020

COUNTRY: Norway

FUNCTION: Drilling and Completion Operations

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During a nightly stock check it was noticed that it had dropped in level. Upon investigation the effluent operator noticed oil in the tank bund and the Tank export line work leaking from the end.

WHAT WENT WRONG?:

The event was caused by microbial corrosion of a weld in a dead leg of the common suction from tank.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Follow an approved change management process prior to altering process systems (even if temporary). Could consider this an issue associated with "proper isolation" also.

BARRIERS:

Management System Element Barrier Failure: Policies, standards and objectives

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: Aug 31 2020

COUNTRY: Romania

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

CATEGORIES:

PRD release to atmosphere above threshold in any 1 hour period and results in:

- liquid carryover or
- discharge to a potentially unsafe location or
- onsite shelter in place or
- public protective measure (e.g. road closure)

INCIDENT DESCRIPTION:

On 31.08.2020 around 16:50 the operator from the station noticed a decrease of gas flow in the station. The 8 5/8" gas pipeline from Compressors-SUG was sectioned by a third party excavator while executing work to install communication cables.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

1. Direct causes (substandard acts and conditions)
 - Gas pipeline damaged by a third party
2. Underlying causes (contributing factors / pre-conditions)
 - Third party company performed unauthorized work around gas pipeline
 - Third party company did not requested authorization from authorities
 - Third party company considered no authorization needed for such work
 - Third party took the risk of unauthorized work based on experience of such work performed in the past
 - Third party company was not aware about pipeline markings
 - Operating company was not aware about work performed around gas pipeline - Third party company did not requested approval from Operating company
3. Root causes
 - Third party did not comply with legal requirements regarding performing works around oil and gas facilities

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Reporting of any activity performed by third-party individuals and companies in the vicinity of the company's facilities and pipelines is important to prevent similar incidents. Increase personnel awareness and update the training on emergency shut down of oil and gas production facilities and pipelines.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate communication

TIER 1 PROCESS SAFETY EVENTS

EUROPE OFFSHORE

DATE: Dec 8 2020

COUNTRY: Denmark

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Emergency shutdown

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

PRD release to atmosphere above threshold in any 1 hour period and results in:

- liquid carryover or
- discharge to a potentially unsafe location or
- onsite shelter in place or
- public protective measure (e.g. road closure)

INCIDENT DESCRIPTION:

Due to an H2S alarm on the inlet ducting to the main generator, a manual ESD2 was initiated. During the depressurization of the plant, the gas lift supply ESDV failed to close fully allowing gas to flow through the vent header and out via the vent stack.

WHAT WENT WRONG?:

1. Technical failure: Failure of actuator of the valve. Alarm received but not acknowledged
2. Organizational failures (operating and control not updated) 3. Design failure

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Alarm rationalization study to be carried out to identify the highly critical alarms and shown separately on the HMI.

BARRIERS:

Management System Element Barrier Failure: Organization, resources and capability

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

TIER 1 PROCESS SAFETY EVENTS

DATE: Dec 9 2020

COUNTRY: Norway

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other

POINT OF RELEASE: Subsea: Subsea equipment

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Subsea leak of methanol when wells were shut in during pressure rate test. During the test and well shut in there was a methanol leak to sea, which was only discovered during a new methanol pumping operation 8 hours later.

WHAT WENT WRONG?:

Nonadherence and miscommunication of operating procedures that has led to several methanol spill to sea incidents. Approval of planned tests and programs not done with all shift teams.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The implementation and follow-up of operating procedure should be reviewed as it stands out as a failed barrier because it was not followed per the intention. The barrier that worked was that the operations team identified the failed barrier when they resumed running as per procedure.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/Organizational culture

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

TIER 1 PROCESS SAFETY EVENTS

DATE: Feb 18 2020

COUNTRY: UK

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Upset

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

Catastrophic turbo failure resulting in a fire within the exhaust of the engine.

WHAT WENT WRONG?:

High temperatures and backpressure in the exhaust of Aux. Gen. B due to a blocked catalytic converter caused one of the turbos in the engine to catastrophically fail.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review the equipment boundaries for key Safety Critical Packages and ensure that ancillary equipment has the same safety critical classification. This should include, but not be limited to, exhaust systems and condition monitoring equipment. Review Exceptions Registers and confirm that abnormal results from condition monitoring are being actioned with the appropriate priority and that the consequence of continuing to operate with the abnormal condition are understood.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS [CONDITIONS]: Protective Systems: Inadequate/defective guards or protective barriers

TIER 1 PROCESS SAFETY EVENTS

MIDDLE EAST ONSHORE

DATE: Feb 7 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 6" aboveground line coming from manifold and going to gathering centre. The line was isolated to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Corrosion/ erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Feb 20 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 6" underground line coming from well and going to Gathering Centre. The line was isolated to arrest the leak.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

Underground flow line leak - Corrosion/ erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Mar 8 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Sump/pit overflow

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude Oil spill from BOP pit of well which belongs to a Gathering Centre. The pit valve was isolated. The leak was vacuumed.

WHAT WENT WRONG?:

BOP valve was slightly open.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Verify all the valves and conditions near the well.

BARRIERS:

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: May 28 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 6" aboveground Transit line-2 coming from manifold and going to gathering centre. The line was isolated to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Corrosion/ erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Jun 13 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 6" aboveground line coming from oil well and going to gathering centre. The line was isolated to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Corrosion/ erosion

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Jul 8 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude Oil leak from 6" aboveground line coming from manifold and going to gathering center. The line was isolated to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Corrosion/erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Aug 11 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 6" underground flow line coming from oil wells and going to Gathering Centre. The line was isolated to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Underground flow line leak
- Corrosion/ erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Sep 3 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 6" aboveground flow line coming from Well and going to Gathering Centre. The line was isolated to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Corrosion/erosion

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Sep 24 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 30" underground transit line coming from a manifold. The line was clamped to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Corrosion/ erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Oct 2 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Drain

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude Oil leak occurred from 2" drain line of the transit pump. The line was isolated to arrest the leak.

WHAT WENT WRONG?:

Drain line leak - Corrosion/erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Dec 4 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 30" underground line coming from Manifold and going to EPF (Early Production Facility). The line was isolated to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Corrosion/ erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing.

DATE: Dec 7 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 6" underground flow line coming from well and going to Gathering Centre. The line was isolated to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Corrosion/ erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Dec 8 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 6" underground flow line coming from Oil Well and going to the Gathering Centre. The line was isolated to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Corrosion/ erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Sep 19 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 6" aboveground flow line coming from Well and going to Gathering Centre. The line was isolated to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Corrosion/ erosion

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Dec 9 2020

COUNTRY: Kuwait

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 30" underground line coming from manifold and going to EPF (Early Production Facility). The line was isolated to arrest the leak.

WHAT WENT WRONG?:

- Wear and Tear
- Corrosion/ erosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Jun 11 2020

COUNTRY: Kuwait

FUNCTION: Drilling and Completion Operations

NUMBER DEATHS: 2

ACTIVITY: Drilling

MODE OF OPERATION: Unspecified

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

- Rig Crew started gas bleeding at 02:15 hrs. from well casing to the Far Pit.
- After finishing gas bleeding, while preparing to kill the well, a flash fire occurred at the pit, which caused severe burn injuries to two Roustabouts stationed near the pit. The fire was extinguished on casing valve closure.
- Both injured persons (IPs) were transferred to Hospital, where the first IP died after few hours on the date of incident, whereas the 2nd IP died after 3 days.

WHAT WENT WRONG?:

- A sudden high velocity fluid surge from the well.
- Fire in the pit likely from ignition due to friction in the pit or probable static electric discharge between gas and the pit discharge line.
- Misperception of risk by crew.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- No crew is to be allowed near flare pit during bleed off operations.
- Ensure periodic inspection and maintenance of adequately designed grounding/earthing for effective dissipation of static electricity accumulated on the vent/drain pipe up to the pit.
- Enhancement of awareness about static electricity hazards in flammable atmospheres.
- Assess and manage Covid-19 impact on monitoring as well as execution of critical activities.

BARRIERS:

Hardware Barrier Failures: Ignition Control

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

TIER 1 PROCESS SAFETY EVENTS

DATE: May 21 2020

COUNTRY: Oman

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leaked from Flow line (9.5 bbls). (Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Apr 1 2020

COUNTRY: Oman

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oily water leaked from remote manifold 4" trunk line, caused by a Pinhole on the line.

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Jun 2 2020
COUNTRY: Oman
FUNCTION: Production
NUMBER DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

INCIDENT DESCRIPTION:

Oil leaked from Flow line (9.5 bbls). (Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management

BARRIERS: No Barriers Allocated

CAUSAL FACTORS: No Causal Factors Allocated

DATE: Jul 5 2020
COUNTRY: Oman
FUNCTION: Production
NUMBER DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leaked from Flow line (10 bbls). (Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Jul 14 2020

COUNTRY: Oman

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oily water leak from flow line (27 bbls). (Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on flow lines and proper Corrosion management.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Sep 2 2020

COUNTRY: Oman

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leaked from 4" trunk line (32.5 bbls). (Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Sep 6 2020

COUNTRY: Oman

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leaked from 4" trunk line (Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Oct 1 2020

COUNTRY: Oman

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leaked from 4" trunk line (Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Oct 11 2020

COUNTRY: Oman

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leaked from flow line (Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Oct 15 2020

COUNTRY: Oman

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leaked from 4" trunk line (Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Oct 16 2020

COUNTRY: Oman

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leaked from 4" trunk line(Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Nov 3 2020

COUNTRY: Oman

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leaked from 4" flow line . (Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Dec 21 2020

COUNTRY: Oman

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leaked from 4" trunk line (Pinhole leak)

WHAT WENT WRONG?:

Pinhole developed on the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Frequent checks on lines and proper Corrosion management.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Apr 22 2020

COUNTRY: Qatar

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Start-up

POINT OF RELEASE: Equipment: Heat exchanger

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During Train-1 start-up, Panel operator received several combustible gas alarms in Train-1 at Unit-5 and Unit-6. Upon operator notification and surveillance, it was identified that gas leak was from newly installed 01-E0702 heat exchanger (De-ethanizer Re-boiler). The leak source was from bottom channel head (between 2 bolts).

Gas leak has approached nearby units triggering the necessity for Train-1 and Train-2 shutdown (as a precaution) and subsequent depressurization.

WHAT WENT WRONG?:

Incorrect gasket was installed between shell and channel head assembly by OEM (Original Equipment Manufacturer) that failed ultimately during start-up.

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Continuous operator surveillance during train start up is crucial to identify any abnormalities and leaks.
2. Quick emergency response and timely decision to shut down adjacent train prevented potential escalation of event.
3. Gasket and bolt verification shall be a mandatory stage for TPI (Third Party Inspection) before releasing the equipment for pressure testing.
4. Always follow procedure requirements and refer to the approved / last revision drawings while preparing gasket list and equipment assembly.
5. In case of any major scope changes, detailed review of all deliverables shall be conducted and reported to company for approval.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

MIDDLE EAST OFFSHORE

No Tier 1 PSE narrative reports were provided

TIER 1 PROCESS SAFETY EVENTS

NORTH AMERICA ONSHORE

DATE: Jan 31 2020

COUNTRY: Canada

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

On Jan 31st at approximately 1:30am a failed SRU boot solvent-water interface shutdown allowed solvent to enter a produced water distribution header which had unknowingly previously frozen leading to est. 750kg release from 3 locations. The rapid loss of containment led to several gas detectors exceeding 30% LEL, which automatically triggered the PAGA (leading to a safe evacuation & no injuries), and automatically shut down the plant and isolated the damaged system limiting the total release.

WHAT WENT WRONG?:

- Improper temperature sensor placement causing electric heat trace (EHT) circuit to not be active as temperature set point was met.
- No electric heat trace (EHT) review of stagnant lines.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Implement program for dead leg detection and proactive management of freezing risks.
- Temperature sensor placement review for EHT circuits with multiple flow paths in froth area using EHT P&IDs. Prioritize lines that could lead to solvent release and lines that could be stagnant.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Jul 31 2020

COUNTRY: Canada

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Fired heater/Boiler/Furnace

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

- Rapid combustion event followed by a fire which was contained to the salt bath heater building. The plant was immediately evacuated and facility shutdown. No personnel were injured.
- The failure resulted in a release of gas, which ignited, causing the explosion. The fire was fuelled by produced gas as the plant was depressurizing to flare. The fire was put out roughly 10hrs after the initiating event.

WHAT WENT WRONG?:

- The regen gas process coil failed at the last bend prior to exiting the salt bath. Failure was due to wall thinning caused by sulphidation and localized erosion-corrosion. Wall loss accelerated over the life of the asset due to H₂S increase. The process coil was in service for 32 years prior to failure. Sulphidation is corrosion resulting from reaction with sulphur compounds (H₂S) in high temperature environments (usually starts >260C).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Equipment End of Life (EOL) assessment:
 - Ensure EOL assessments use correct inputs for both design conditions and inspection results.
 - Ensure EOL assessment is completed by qualified personnel and peer reviewed.
- Equipment strategies:
 - Review Equipment Strategies at regular intervals while considering changes in process conditions.
 - Ensure inspection instructions are clear and bend extrados are measured when erosion-corrosion is possible.
 - Ensure that third party inspection reports are reviewed promptly for important findings as well as accuracy.
- Recognition of Facility Integrity Risk:
 - Ensure that the risks of Sulphidation, even at moderate levels of H₂S, are recognized and mitigated. Sulphidation can occur at levels below "sour service" limits and with materials suitable for sour service.
 - Even when operating conditions slowly change over time, like H₂S creep, ensure that proper Risk Assessments and Management of Change are followed.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

TIER 1 PROCESS SAFETY EVENTS

DATE: Mar 22 2020

COUNTRY: Canada

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

At 3:29pm LEL in the Compressor building starts to rise. One minute later LEL hits 20% and the Board Operator stops the compressor. By 3:31pm LEL reaches 43% and the Unit Operator witnesses doors blow open on the Compressor Building. A second Unit Operator then observes a small fire on top of the engine near the turbo side. At 3:38pm the fire eyes malfunctioned.

WHAT WENT WRONG?:

Fuel Gas Filter leaked. Operations stopped compressor but did not hit USD and did not isolate Fuel Gas supply until 7 minutes later. Compressor fuel supply not equipped with an adequate ESDV and not designed to isolate on 40% LEL in the compressor building.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Develop an EGP specific fire and gas detection response philosophy (based on the way the system is design today) including specific guidance on what to do on 20% LEL, 40% LEL, H2S, and Fire. Make this a safety critical document, provide training and add to competency matrix. Work with Maintenance and F&C to develop a reference list or diagram of all sources of hydrocarbons and H2S flowing into each building along with the location/tag of the valve that needs to be closed to isolate the sources. Issue the list or diagram in a training bulletin as an interim measure, until the action item to implement a fire and gas upgrade project is completed. Complete the conceptualization phase for the fire and gas upgrade project internally, including a gap analysis of current state against Repsol standard (this can be done in conjunction with developing the interim reference list for operations). Review the risk ranking and update if required. Develop an approval memo for submission into the 2021 budget with additional detail and a Class 5 cost estimate to complete the upgrades.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

TIER 1 PROCESS SAFETY EVENTS

DATE: Sep 29 2020

COUNTRY: Canada

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

Fire Tube Failure resulting in a Fire. Liquid carry over on the flare stack resulting in ground fire.

WHAT WENT WRONG?:

Root cause(s) Maintenance management

- Inspection plan does not exist or is not implemented Root cause(s) Maintenance management
- Maintenance plan does not exist or is not implemented Root cause(s) Procedures and control of operations
- Lack of procedure

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Generate Fire Emergency response instructions for the 12-19 Battery prior to Battery start up. Generate Emergency response instructions for all other batteries in the asset for responding to abnormal (dark/black) smoke or fire from burner stacks in fire tube vessels. Conduct a field level workshop to determine if there are any additional scenarios where a shutdown procedure is required. Develop the procedure(s) required based on the scenarios found.

Update maintenance PM (beyond current function testing) scope and steps of ESDVs to be in accordance with the requirements of the ESD Systems Performance Standard and manufacturer recommendations (whichever is more stringent). Update Performance Standard as required. Update current Firetube Repair Procedure. Add firetubes to the Pressure Vessel Performance Standard (PC007). Include design and maintenance requirements. Ensure firetubes are captured in both the CMMS and Integrity D.

BARRIERS:

Hardware Barrier Failures: Ignition Control

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

TIER 1 PROCESS SAFETY EVENTS

DATE: Jan 16 2020

COUNTRY: Canada

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

On January 16th there was an explosion and subsequent flash fire that occurred around the Fuel Gas Letdown buildings. At approximately 1:15am based on DCS information LEL levels started to rise and continued to do so unnoticed until the call centre notified the operator that 20%+ LEL existed inside the building. Operators were unaware of the levels as LEL set points are set at 20% low level and before action could be taken the explosion occurred.

There were no injuries.

WHAT WENT WRONG?:

Given the extreme cold, the light ends (C3+) in the relieved gas stream condensed in the vent header and drain out of sleeve connections, and rainout of vent tip about 40 minutes outside of the buildings associated with the co-gen facility buildings.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Changes to be made to DCS to divert compressors into re-injection mode as result of any upset in refrigeration system.
2. Train all operators on changes to system.
3. Changes to plant fire & gas system to alarm at 10% LEL and high level at 20%.
4. Add external audible and visual LEL alarms.
5. Car seal close manual bypass valve to prevent gas blow by.

BARRIERS:

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

TIER 1 PROCESS SAFETY EVENTS

DATE: Jan 21 2020

COUNTRY: Canada

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Upon entering the lease, the operator could see and hear gas venting from the water production tank. The Operator shut their truck off and followed the lease/building entry procedures. Once inside the well building, the operator found the L2 level controller for the produced water dump was stuck open, which allowed gas to vent to the produced water tank. The operator proceeded to isolate the ball valve downstream of the dump valve to stop the flow of gas to the tank.

WHAT WENT WRONG?:

1. Repetitive failure of produced water dump valves.
2. Inadequate level control in upstream separator.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. High level shutdown functionality on separator and auto-dump on fuel gas scrubber, These have been incorporated into new designs .
2. Source acceptable replacement for floats in the fuel gas scrubbers at problematic well locations.
3. Determine feasibility of moving fuel gas take off point, away from condensate re-injection point at wells that have chronic fuel gas flooding issues.
4. Determine feasibility of adding:
 - auto dumps to the fuel gas systems on older wells that have had or the potential to have fuel gas flooding issues
 - a low-level switch to well site older vintage separators
 - an in-line water meter to detect a gas flow rate and ESD well locations
 - an in-line water meter that is monitored and alerts operations to metering inconsistencies/ issues
5. Operations to build a list of problematic wells that require more frequent visits, or unusual characteristics.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

TIER 1 PROCESS SAFETY EVENTS

DATE: Oct 20 2020

COUNTRY: Canada

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

An operator completing their rounds entered the inlet header building. Upon entering they noticed the oil transfer pump was leaking oil from the seal area. There are two transfer pumps operating in parallel in this service, the leaking pump was isolated to stop the leak. The second pump was inspected to ensure its seals were sound and fit for service. Operation continued with just the single pump running, and spill clean-up was completed. Approximately 1.1 cubic metre of produced oil was released into the building sump containment under grating. Initial pump repairs identified issues with O-ring material.

WHAT WENT WRONG?:

1. The pump had incorrect O-ring installed.
2. O-Ring Materials & Specifications for the Transfer Pumps not outlined for multi service pumps.
3. Maintenance histories show repeat failures, no action taken.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Maintenance is gathering a district pump list that will be risk ranked on potential process safety event classification criteria and leak severity.
2. Additional alarms and shutdowns will be evaluated for installation to limit the release volume and improve the response time for operations.
3. Reduce reliance on operator rounds as a means for identification of leaks.
4. Maintenance will update the task list with these pump work orders to identify the proper equipment to be used.
5. Increased awareness, education and training with field operations on reporting requirements for incidents relating to process safety events.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

TIER 1 PROCESS SAFETY EVENTS

DATE: Nov 11 2020

COUNTRY: Canada

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

While completing daily morning production report and checking well status from control room. The operator noticed a well had an ESD on high LEL and that methanol tank level was very low. Suspecting a possible methanol leak the operator travelled directly to site. Upon arrival methanol was found dripping out of door and overflowing onto the ground under building. The source of the leak was due to the packing nut and packing itself were pushed completely out of the fluid head body.

WHAT WENT WRONG?:

1. Pump had been replaced the day prior and put back in service.
2. Packing nut appeared to have “backed off” threads.
3. Methanol was gravity fed from tank and leaked from housing.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Verify packing nuts are secured - create procedural document.
2. Share learnings to other areas to prevent similar incidents.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

TIER 1 PROCESS SAFETY EVENTS

DATE: Oct 22 2020

COUNTRY: Canada

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Operations responded following proper protocol for a building gas leak at the CS Pad and discovered the gas compressor leaking. Upon further investigation, it was discovered that the 2nd stage compressor failed leading to the compressor casing damage. There was notable internal damage to the 2nd stage piston was out of the cylinder. It was also confirmed that both Lower Explosive Limit (LEL) sensors had exceeded 100%.

WHAT WENT WRONG?:

Compressor crankshaft failed as the result of a one-time brittle overload fracture.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Conduct a formal Hazop/LOPA analysis of the CS-Pad facility ensuring that catastrophic compressor failure is considered. Potential mitigation strategies may include:

- Installation of automatic isolation valves on the compressor gas supply.
- Installation of a vibration transmitter on the compressor frame with the capability to shut down the compressor on high vibrations.
- Creation of an alarm to notify operations if a well test compressor runs for an extended period.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

TIER 1 PROCESS SAFETY EVENTS

DATE: Mar 1 2020

COUNTRY: Canada

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Operator walking by the building noticed a hydrocarbon odour. On investigation Operator witnessed fluids coming from the pump body releasing to the building floor.

WHAT WENT WRONG?:

Suitability of pump for service application requires review; assess level switch in building floor sump; Reassess set points for LEL and H2S monitoring in building.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Know the condition of your safety devices. Risk assess any impairments or deferrals.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

DATE: Aug 27 2020

COUNTRY: Mexico

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Early morning, a services company supervisor notified production department about an oil spill from the frac Tank installed in the well (12 hrs. after it had been installed). Production Volumes were being measured in the Frac Tank every hour by the operator, not realizing the speed of volume produced was going to fill the tank in next hour.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

1. Hazard Identification & Risk Control
2. Competency & Training
3. Contractor Management

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Contractor Management: Update and review of Contractor Management Procedure;
2. Review Hazard Identification & Risk Control Procedure for Contractor be engaged.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: Aug 11 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Platform/Well Pad Flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Gas was releasing from the production tree cap. Well was manually isolated and the release was stopped. incident investigation has been initiated.

WHAT WENT WRONG?:

MOC process didn't highlight the inclusion of the weep hole and O-ring decompression could cause failure. Operations were not aware the weep hole exist till the event happened. Design of production tree did not allow for identifying a leak through weep hole. Swabbing procedure was not sufficiently detailed to highlight and manage the risk.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Failure of the O-ring was due to explosive decompression (ED), not due to material compatibility. The Ultimate cause for the ED damage likely relates to cyclic loading, pressure fluctuations encountered during shutdown and restarting of the system. Remove tree caps with weep hole and O-ring. Refine swabbing procedure O-ring inspection. Monitor well pressure during real time.

TIER 1 PROCESS SAFETY EVENTS

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: Jul 22 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Liquid release from the flowline. 2 holes were found as release points.

WHAT WENT WRONG?:

CO2 Corrosion in the flowline. Failure to identify hazard or risk within equipment design. Lack of Preventive maintenance (Chemical Program), lack of Monitoring/Warning Systems (Pressure Monitoring Equipment).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Design Modification: The historical design has been modified by the facilities engineering staff, to accommodate for operational conditions and corrosion tolerances specific to the typical shale operational requirements and exposures. This will be reflected within the Piping Standard documents.
- Operational Flowline Assessments: Inspection (smart pigging, ultrasonic wall thickness) is planned of the similar, historically constructed flowlines to proactively identify potential erosion and corrosion issues. Operational flowlines with similar construction parameters have been identified and will be inspected, retrofitted, or replaced to ensure operational soundness.
- Environmental Risk Assessment: Operation HSE will conduct an environmental risk assessment to identify other potentially sensitive areas of operation and communicate with Operations team to ensure proper safeguards are implemented.
- Chemical Program Implementation: Optimization of chemical program utilizing risk-based prioritization approach to identify and mitigate potential corrosion related issues is completed. The chemical program, which includes various corrosion inhibitors, has been implemented throughout all assets, including new wells, flowlines and facilities.
- Pressure Management Equipment: A field level risk assessment of current flowline installations is in the final phases of completion. Once completed, pressure management warning systems, including low pressure monitoring, will be installed where required.

TIER 1 PROCESS SAFETY EVENTS

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Protection Systems - including deluge and fire water systems

Human Barrier Failures: Response to emergencies

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

Management System Element Barrier Failure: Assurance, review and improvement

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Aug 19 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A 1" nipple failure on a rental compressor second stage gas scrubber.

WHAT WENT WRONG?:

Unit continued to run after the 1" nipple failed, the solenoid on suction valve did not close.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

None Identified

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Aug 12 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Gas well shut in on low flowline pressure alarm. Operator discovered a 3/8" diameter hole on the bottom of intermitter valve body, resulting in gas release.

WHAT WENT WRONG?:

Hole in valve body from internal erosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

None identified.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Jul 2 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Dump valve body cut out from internal erosion and released gas.

WHAT WENT WRONG?:

Product flow eroded the valve body.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

None identified.

TIER 1 PROCESS SAFETY EVENTS

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

DATE: May 22 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Failure of threads of a concentric reducer (3" x 2") on the inlet piping to a rental compressor caused a release of natural gas.

WHAT WENT WRONG?:

Threads on concentric reducer failed.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

None identified.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Apr 3 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A water hauler was attempting to transfer condensate. The 4" offload valve was frozen, so the contractor hauler shifted to use the 2- inch valve on the bottom of the tank. As the contract operator turned the bull plug to remove it the 2- inch tailpiece portion of the valve unthreaded. Once the 2- inch tailpiece became unthreaded the interior ball of the valve fell out of the plug and condensate began to uncontrollably flowing from the valve. This resulted in the release of 170 barrels of condensate and 20 barrels of produced water into secondary containment.

WHAT WENT WRONG?:

The internal ball fell out of the valve body allowing fluid in the tank to flow uncontrolled into the secondary containment.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Perform an inventory of 4" and 2" header/loader tank valves and develop a plan to mitigate unintentional unthreading. Update offloading procedure to designate which valves to utilize and when to stop the job.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: Mar 2 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Start-up

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A 1" pipe nipple on a compressor unit failed. The nipple was located on a hot gas bypass connection on the compressor discharge between the third stage cylinder and final gas cooler. No injuries resulted from this incident.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

The compressor unit is started and the compressor unit 3rd stage piping begins to vibrate at the hot gas bypass piping connection.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Identify vibration points and eliminate or install supports.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

DATE: Mar 2 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Condensate was released during a tank overflow.

WHAT WENT WRONG?:

Multiple failures occurred leading to the incident:

1. level transmitter in tank failed to report, allowing tank to overfill without shutting in wells;
2. poor liquids separation was occurring at the high pressure separator oil dump valve sending oil and water to the oil tank; and
3. only one oil inlet valve was open, sending all liquids to oil tank. All condensate remained in secondary containment with slight residual spray which did not penetrate the ground.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Ensure transfer procedure are complete and level transmitters are functioning properly.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Detection Systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

TIER 1 PROCESS SAFETY EVENTS

DATE: Feb 26 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

An operator arrived on location and found gas venting out of the gas lift regulator skid. The facility was shut down and leak isolated. The incident volume is 0.148 MCF of natural gas.

WHAT WENT WRONG?:

1" pipe had broken causing the gas leak, due to extreme vibration coming from the regulators.

Common causes: Clarity of signs, signals, instructions and other information; Equipment Design.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Design and inspections issues. Looking into wider usage of remote vibration monitoring.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Jun 26 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)

CATEGORIES:

PRD release to atmosphere above threshold in any 1 hour period and results in:

- liquid carryover or
- discharge to a potentially unsafe location or
- onsite shelter in place or
- public protective measure (e.g. road closure)

TIER 1 PROCESS SAFETY EVENTS

INCIDENT DESCRIPTION:

An incident with a release of 4.205 bbls of oil. An area of 225 ft x 150 ft by .08 inches in depth. The gas V-ball that maintains the set pressure on the separators failed to operate, causing the separators to go into a high psi. There was a command for the inlet valves to close, but the inlet valve to the west separator did not close. Since the inlet valve did not close the vessels pressured up and caused the relief valves to pop off on both vessels, at the same time. PSV's have a 1.25" diameter orifice, and are set for 120psi.

WHAT WENT WRONG?:

Process Emergency Shutdown Valves (ESDV's)

Common Cause - work pressure production vs. safety (PM frequency of ESDVs reduced).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Know the condition of your safety devices. Risk assess any impairments or deferrals.

Seven actions, addressing drier system, line drainage, replace inlet ESDV, retest PSVs, evaluate risk based approach to ESDV PMs.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Jul 9 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Overflow on water tanks due to failure of high level ESD. 3240 bbls of water and 35bbls of oil released, all recovered in lined containment.

WHAT WENT WRONG?:

Management of Change Procedures.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Follow an approved change management process prior to altering process systems (even if temporary).

Ten actions, covering SOPs for water transfer facilities, alarm management, remote shutdown capacity, validation of programming, new systems checks, expectations around function testing.

BARRIERS:

Management System Element Barrier Failure: Policies, standards and objectives

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

TIER 1 PROCESS SAFETY EVENTS

DATE: Aug 26 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

MSO received notification from DOC that SCADA was not registering tank levels. Upon arrival MSO found an overflow event, started pumps in hand to stop the release, and isolated the facility. Approximately 521 bbls of mixed oil and produced water (106 bbl oil and 415 bbls produced water) were released and stayed within the secondary containment. Work orders have been submitted for remediation and repairs. Leadership was notified and CM was contacted.

WHAT WENT WRONG?:

Lack of robust procedures for recovering from system faults including loss of communication to remote controls. Several contributing factors identified, including MOC not followed.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Eleven actions, focusing on systems changes, new procedures (SOPs), MOC adherence, 3rd party review of Delaware control systems.

Follow an approved change management process prior to altering process systems (even if temporary).

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: Aug 30 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other

POINT OF RELEASE: Equipment: Pig launcher/receiver

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

While removing a foam PIG and a sizing PIG from a pipeline that was under construction, the sizing pig ejected from the temporary PIG catcher, as the foam PIG was being taken out, causing the IP's hand to swing back and hit their arm on the temporary PIG catcher open hatch resulting in a LWC and overnight hospitalization.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

1. Equipment/Energy Isolations
2. ITM - Inspection, Testing, and Maintenance
3. Procedures
4. Training

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Four actions:

1. Development of L48 Pigging SOP, including competency requirements with focus on isolation verification
2. Evaluate and implement pig extraction best practices
3. Minimum design and inspection requirements for temporary pigging equipment to be developed
4. Pigging activities to be classed as a critical activity ensuring risks are managed

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Policies, standards and objectives

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

DATE: Sep 8 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Upset

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Inlet valve failed allowing LOPC. Investigation on going.

WHAT WENT WRONG?:

ESD failed to actuate when command was given. The air solenoid relieved air pressure which should have actuated the valve into the closed position. Upon inspection of the ESD Valve, large amounts of corrosion were discovered in the actuator verifying that maintenance had not been performed on the valve in some time. No PM for this equipment could be identified other than 5yr facility checklist.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Four actions - Evaluate PM for critical valves, Refresh LOTO policy, Evaluate wireless telemetry (ESD if comms are lost), adjust pressure transducer to show total liquid rather than total water.

TIER 1 PROCESS SAFETY EVENTS

BARRIERS:

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Nov 26 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Start-up

POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

While attempting to circulate through the H₂S side and dropping into the CO₂ side of T3 process, a rupture disk burst. A large volume of gas was released into the process building via a missing plug between the rupture disk and the PSV.

WHAT WENT WRONG?:

Ruptured disk.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

"Walk The Line. Know the condition of your safety devices."

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

TIER 1 PROCESS SAFETY EVENTS

DATE: Nov 12 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Unspecified - other

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

Production Operator doing routine surveillance discovered a fire on the gas lift compressor skid of a remote well pad. Operator activated ESD shutting in the wells and isolated the pipeline. Emergency services extinguished the fire.

WHAT WENT WRONG?:

Sensing line leak due to improper installation and vibration. The flow from the leak was deemed low, but the resulting fire escalated to release lube oil and other liquids which fed the fire and aided in causing the final extent of damage.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Vibration related failure on small bore piping due to installation.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Human Barrier Failures: Response to emergencies

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

TIER 1 PROCESS SAFETY EVENTS

DATE: Oct 29 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

It was observed that the Pressure Relief Valve (PRV) exhaust pipe was loosed from the compressor.

The compressor shuts-in but suction pressure could flow.

WHAT WENT WRONG?:

Excessive vibrations.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

DATE: Jan 15 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

- On January 15th at 22:35, a trouble alarm came in on 'C' Recycle compressor; followed by failure of the recycle compressor resulting in a CO₂ release inside the recycle compressor building
- All five building CO₂ detectors maxed out at 20,000 ppm

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

- Lab concluded studs failed through a combination of fatigue and overload failure as applied stress increased on un-cracked studs
- Most likely cause of bolt fatigue was misalignment of cylinder or inadequate torque of studs
- Existing compressor monitoring/alarm system not capable of providing early warning of failure

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Validate fit-for-service status of cylinder to distance piece studs and supports
- Retain controlled access to the building under SCBA per SBRA until fit-for-service validation is complete on the four identical recycle machines currently in operation
- Update processes and procedures for cylinder alignment, surveillance and maintenance
- Long term - review cylinder support design and evaluate monitoring/control system upgrades

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Mar 28 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Wells, drilling and intervention: Well

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

- H2S leak from a transmitter box (193PT9610) on well was detected by nearby detector triggering toxic gas alarm
- DCS high H2S ppm on wellhead at the tree was 246 ppm
- Operators isolated PT9610 by closing needle valve upwind of PT enclosure
- No personnel were at site during the release

WHAT WENT WRONG?:

Inadequate gasket material.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Graph foil gaskets to be replaced with Teflon Gaskets as found.

TIER 1 PROCESS SAFETY EVENTS

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

DATE: Aug 19 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Workover / Well services

MODE OF OPERATION: Well intervention / Well servicing

POINT OF RELEASE: Wells, drilling and intervention: Well

INCIDENT DESCRIPTION:

- Coiled tubing (CT) was utilized to perform an acid stimulation job on new perforations and directly following the same CT was used to lift the well to production with Nitrogen.
- While pulling the CT out of the well at 400 ft. from the surface, the tubing separated at the injector head and BOP area.
- A 30 ft. tubing section of approximately 20 lbs. was thrown approximately 100 ft. away from the well location and approximately 18,200 lbs of nitrogen were released from the well.
- No H₂S was detected by fixed nor by personal monitors.

WHAT WENT WRONG?:

- Investigation and lab analysis determined a larger volume of unspent acid (different than previous jobs), in combination with well gas (H₂S and CO₂) resulted in exponential increase in sour corrosion rates of the CT. Incremental adjustments (planning and operational) resulted in larger volume of unspent acid available to react with well gas for a longer period of "contact time", in the wellbore environment.
- Additionally, while lifting the well with nitrogen (when coiled tubing was at a depth below the perforations), the coiled tubing was exposed to turbulent flow of spent/unspent acid and well gas from the formation. In turbulent flow, the inhibition effectiveness of the corrosion inhibitor in the acid can be disrupted and may have also contributed to the corrosion of the CT.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Pre-job design did not identify all required inhibition scenarios.

- Design included inhibition requirements for HCl and well gas separately.
- Design did not account for environment created when large quantities of acid and well gas are in combination.
- Design not fully inclusive of inhibition requirements during both well stimulation and well flowback operations.
- Design and pre-job reviews did not identify how exposure of coiled tubing to turbulent flow from perforations can diminish corrosion inhibitor effectiveness.

Well Procedure contained steps and terminology that were not clear and/or were not followed resulting in increased length of time CT was exposed to combined acid and well gas environment.

BARRIERS: No Barriers Allocated

CAUSAL FACTORS: No Causal Factors Allocated

TIER 1 PROCESS SAFETY EVENTS

DATE: Oct 14 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

- H2S detected inside the Pro 2 building resulting in gas alarms in the control room. No personnel exposed
- Personnel proceeded to investigate using supplied air. H2S detectors indicated concentrations up to 60 ppm at the building mezzanine walkway
- Despite multiple attempts personnel could not locate the leak and decided to initiate a controlled shutdown of the Pro 2
- After purging with nitrogen, located a 1/8" hole at the LP Flash PSV inlet line (carbon steel line)
- The leak resulted in a Tier 1 based on the hole size and leak duration greater than 1 hr.

WHAT WENT WRONG?:

- Leak occurred in a relatively long and stagnant pipe section. Line identified as "Do No Pocket" in the P&ID but there is no slope defined
- Post-release inspection revealed loss of wall thickness and evidence of liquid accumulation in the vicinity of the hole
- Similar failure occurred in upstream pipe. Metallurgy replaced with stainless steel in both trains

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Inspect other lines potentially subject to similar issues
- For future projects, consider that requirement to "Do Not Pocket" may not be sufficient if the line could sag between supports accumulating liquids, consider defining a slope requirement or more resilient metallurgy based on cost-benefit

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

TIER 1 PROCESS SAFETY EVENTS

DATE: Jan 15 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The procedure to remove a blind flange was followed. Due to sedimentary build up that plugged the drain, all fluid was not able to freely drain from the separator. After removing the flange, approximately 80 bbls of oil discharged from the separator (FWKO). 79 bbls was recovered.

WHAT WENT WRONG?:

Construction/Installation Less than Adequate.

Attention to Detail.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Improper valve handle installation and assumption of valve positioning allowed for fluid release.
- Distribute field learning focused on LMRA and framing/confirmation bias prior to breaking containment
- Incorporate learning regarding verifying valve positions into the PSSR checklist and review process
- Conduct audit of PSSR process to identify gaps
- Utilize corrected PSSR process before bringing on new equipment

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Jan 20 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Lease operator was prepping FWKO, emptying fluids and de-pressuring vessel, for cleaning when he noticed oil coming out the thief hatch on the skim tank. He immediately stopped emptying the FWKO and by-passed skim tank.

WHAT WENT WRONG?:

- Lack of Understanding of Potential Incident Scenarios (Process Critical Parameters) and Other Risk Analysis
Less than Adequate
- Lack of Hazard Recognition
- Procedure not required but should be

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Do NOT purge fluid/gas into a full tank – Find alternative method to move the fluid.
- If a change of process flow occurs, operations team MUST not leave equipment unattended.
- If a controller is overridden by a human, the human becomes the controller.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: Jan 28 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

6" Low pressure trunk line froze and split, releasing 35 KCF/per hour.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

Line split due to a freeze in the line

CORRECTIVE ACTIONS & RECOMMENDATIONS: n/a

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

DATE: Feb 3 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A SWD received 3rd party volumes unexpectedly, which cause increased tank volumes that spilled over before transportation arrived to collect.

WHAT WENT WRONG?:

- Inadequate Communication between Company and Contractors
- Inadequate Communication between Work Groups / Multi Company Operations - Simultaneous Operations
- Monitoring and Response Less than Adequate

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Develop a line of communication with inter-company production personnel regarding PW operations at the SWD
- Coaching on Stop Work Authority and leaving the location without notification
- Repair the containment liner and identify and repair other damaged liners

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate communication

TIER 1 PROCESS SAFETY EVENTS

DATE: Feb 25 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Skim tank ran over at the Tank Battery due to water dump on heater hanging open and sending all production to skim tank. RRC Spill.

WHAT WENT WRONG?:

Equipment reliability less than adequate.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Feb 13 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

602 kcf released/0 recovered fluid in pipeline/mechanical failure on compressor resulting in cylinder cap coming off releasing gas. Lease operator isolated downed compressor when arriving on location.

WHAT WENT WRONG?:

Fluid / Mechanical Failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Feb 22 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

Agency Reportable. Fire. Fire ignited on the #4 Compressor. Fire Department was called out and extinguished the fire.

WHAT WENT WRONG?:

- Construction/Installation Less than Adequate
- Lack of Quality Assurance/Quality Control

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Enhance PSSE process (e.g., include inspection of connections prior to equipment start-ups, check connections via FLIR during purging, etc.)
- Add clear/markings signage of ESD locations
- Improve MOC process regarding repurposed equipment
- Revisit potential hazard & risks associated with manual isolation and blowdown via risk assessment

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Feb 28 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Agency Reportable. Spilled 20 BO. Recovered 20 BO. Third party roustabout crew was cleaning out oil tanks. They utilized LOTO but missed one valve on the overflow line. The LO opened valve due to high tank levels which caused oil to leak out the open manway.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

Lack of Hazard Recognition

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Review Energy Isolation expectations with all Maintenance crews
- Review Work Authorization expectations with Maintenance Foreman
- Maintenance Foreman involved with this incident will lead the sharing of Incident Learnings with all Maintenance crews
- Share Incident Learnings across the Organization

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: Mar 12 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Water Flood Plant Operator identified a 2" steel CO₂ injection line failure at the well. The release was immediately isolated via lateral line valves from a remote location. Release most likely caused by internal corrosion.

WHAT WENT WRONG?:

External Corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Add more pressure monitoring devices.
- Excavate pipeline to identify potential failure points.

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Mar 14 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The 1" discharge line on a recirculating pump had a tee with a 1x1/4 bushing, the 1/4" nipple threading into the bushing had broken at the threads spraying 30 bbls of oil from the discharge line on the pump at the Central Tank Battery.

WHAT WENT WRONG?:

1/4" nipple sheared off at threads.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Add pump bushing to Maintenance program
- Include guidance to use Schedule 80 or larger fittings on the discharge side of the any pump(s) in spill prevention guidelines

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

DATE: Mar 19 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A high LEL alarm on Compressor Unit B at the Compressor Station. The operator was on site and shut down the unit and discovered that there is a crack on the 2nd stage discharge bottle. The unit has been isolated and the 2nd stage discharge bottle will be replaced.

WHAT WENT WRONG?:

Crack on the 2nd stage discharge bottle.

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Mar 23 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

On 3-23-20 at approximately 4:45 pm the office was notified that a Third-Party Company was laying an optical line and struck a 6" gathering line with 15 to 20 psi and released for approximately 1 hour.

WHAT WENT WRONG?:

Human Error - Third Party Contractor running cable for phone company struck 6" pipeline operating at 15 to 20 PSI.
Task Execution

CORRECTIVE ACTIONS & RECOMMENDATIONS:

n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

DATE: Apr 1 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

While turning well into test, valve was missed causing flowline to pressure up and burst.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

Attention to Detail.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- The valves under the jumper line have been turned to face outward for better visibility and left open, allowing the flow to be controlled via the valves on the header.
- The ESD valves have been properly programmed into the PLC for all wells on the pad.
- ESD function testing will be included in the commissioning process moving forward. Any deviation to allow start-up of the wells with the ESD in manual mode requires production superintendent approval.
- Assure that any deviation in a SOP receives superintendent approval.

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

DATE: May 2 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Pressurized storage vessel

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

ISO tank containing 34.99% Hydrogen Peroxide (HP) relieving pressure through the PRVs and venting a cloud of vapor into the air. After 12 to 15 minutes of relieving pressure, the 47.1lb 22" hatch lid separated from the ISO tank and landed approximately 714' away in the adjacent work area.

WHAT WENT WRONG?:

- Exothermic reaction
- Preventative Maintenance need improvement

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Vendors to develop and distribute procedure for handling HP during bidding process and for existing facilities
- Utilize ISO tanks that have mechanical barriers to prevent back flow and are rated to handle overpressure scenarios
- Ensure ISO tanks are properly vented to prevent pressure build-up

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

TIER 1 PROCESS SAFETY EVENTS

DATE: May 5 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Wells, drilling and intervention: Well

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

37.32 kcf released over a 1.5 hour period. At approximately 8:30 PM on 5/5/20, Company received a call about a gas leak. LO was dispatched to location and found a 1/4" nipple had broken off from the casing (annulus) riser.

WHAT WENT WRONG?:

Equipment failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: May 15 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Upset

POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Central Tank Battery - A hi liquid level alarm went out for the South gas scrubber. At 7:45PM, oil was found coming out of an open 3" blowdown-35.58bbls of oil was spilled. At some point later, the scrubber back-filled through a faulty check valve and a faulty dump valve.

WHAT WENT WRONG?:

- Lack of understanding of Potential Incident Scenarios (Process Critical Parameters) and other risk
- Attention to detail

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Distribute field learning on 'walking the line' to understand impact to process flow upstream / downstream of lock-out/tag-out (LOTO) and how valves may present false zero energy (e.g. dumps, checks, butterfly)
- Update Energy Isolation checklist to address above referenced key field learnings
- Increase field verification on use of LOTO checklists/planning tools (e.g. through Supt site visits, Eng. engagement)
- Incorporate learnings into next revision of Spill Prevention Guidelines and training program (On Site LOTO Procedures topic)

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: May 18 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

6" pipeline rupture releasing 2400KCF natural gas and caused flash fire in remote area (closest public receptor 1.5 miles away). All agencies were notified as required.

WHAT WENT WRONG?:

Equipment Failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Remove any excess bending load that may have been on the pipe
- Hydrotest pipeline for 24 hours to ensure integrity
- Monthly surface-injected batch treatment with corrosion inhibitor and pigging to eliminate corrosion risk
- Assess replacing the 6" line with 8" pipe to allow for pigging

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

TIER 1 PROCESS SAFETY EVENTS

DATE: May 21 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 6:25 AM on 5.21.2020 while checking SCADA Foreman observed low gas suction pressures at the Compressor Stations, Release was due to a tree that impacted and hung up on the pipeline.

WHAT WENT WRONG?:

Weather causing tree to hang up on pipeline.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

DATE: Jun 8 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During routine surveillance the operator discovered oil leaking from an oil storage tank. Oil was contained in firewall.

WHAT WENT WRONG?:

Equipment / Material Failure

Preventative Maintenance

Internal Corrosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

TIER 1 PROCESS SAFETY EVENTS

DATE: Jul 4 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Operations experienced issues with the internal floating roofs (IFRs) of two tanks between 7/04/2020 and 7/09/2020. This event resulted in the internal floating roofs (IFRs) sinking in two oil storage tanks.

WHAT WENT WRONG?:

Equipment / Material Failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Ensure tank analysers are calibrated properly, operating in range, setup to alarms, etc.
- Review tank design and operation procedures

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: Jul 9 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Failure to verify zero energy during valve installation resulted in spill. Vac truck vacuuming during LOTO process. Butterfly valve turned in the wrong direction.

WHAT WENT WRONG?:

- Energy Isolation
- Procedure/Work Practice not Followed

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Upgrade expectations of Energy Isolation as it applies to the process flow. Implement training to deliver the expectations.
- DO NOT have any external media applied to the energy isolation while demonstrating Zero Energy (Pumps, Vac trucks, etc.)
- Calculate the approximated amount of fluid in the flow lines to be worked on. Have an expectation of fluid captured. To be discussed during the planning phase and JSA discussion.
- Train operation team on valve specifications

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: Aug 8 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

Two contract flowback employees heard a pop from the compressor building. Contract employees saw a fire come from the east compressor, tried to extinguish the flames. Left the scene and hit ESD on the way to location entrance. Emergency services were called.

WHAT WENT WRONG?:

Design and Construction.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Evaluate MOC for field changes to vendor packaged equipment
- Evaluate QA/QC process for field modifications and installations
- Monitor for unusual equipment vibration.

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

TIER 1 PROCESS SAFETY EVENTS

DATE: Aug 9 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Platform/Well Pad Flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

8 inch 90 degree fitting failed due to internal corrosion resulting in Produced Nat. Gas loss of approximately 450 KCF. Automated valves and alarms activated as designed limiting loss.

WHAT WENT WRONG?:

Equipment / Material Failure

Preventative Maintenance

Internal Corrosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: Sep 2 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

TRRC reportable spill (100bbls PO 20bbls PW) into an impermeable containment overflowed due to an automated level control switch failure. 120 barrels total of combined liquid overflowed the open top AST into the lined impermeable containment. All fluids were recovered. Investigation ongoing.

WHAT WENT WRONG?:

Equipment / Material Failure

Attention to Detail

Lack of Understanding of Consequences

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Strap storage tanks with color cut once per shift. (Physical level check)
- Ensure a competent operator color cuts and measures fluid levels in tanks.
- Add audible LSHH alarm.
- Additional sight specific equipment and set point changes.

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: Sep 5 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

3/4" valve nipple that is attached to the main suction header failed and was releasing gas. It is estimated that gas released for 1 hour and 45 minutes prior to the operator discovering the issue. 151.5 KCF was released to atmosphere (6,696 Pounds)

WHAT WENT WRONG?:

Equipment / Material Failure

Preventative Maintenance

Internal Corrosion

CORRECTIVE ACTIONS & RECOMMENDATIONS: n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

TIER 1 PROCESS SAFETY EVENTS

DATE: Sep 18 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

L0 arrived at the battery and discovered oil flowing from the bottom of production tank. Upon investigation, a hole was found on the bottom of the tank near the manway. Total Spilled 92 B0, 35 BW, recovered 83 B0 and 24 BW.

WHAT WENT WRONG?:

Equipment / Material Failure

Preventative Maintenance

External Corrosion

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The hole had developed from external corrosion. Operations will install a three-phase separator to transfer fluid and the tank will be removed from service

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: Sep 23 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil Tank overflowed due to dump valve not working properly

WHAT WENT WRONG?:

Equipment / Material Failure

Preventative Maintenance

Monitoring and Response Less than Adequate

In Past no Incident Occurred

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Mitigation: Operator reset the PSI on the level controller and monitored the separator to verify that the dumps were working correctly. After reviewing WINTMC and evaluating the trend the level controller failed to open.

1. Create Incident Learning and distribute to teams
2. Change LSH to call out from text only
3. Change LSH and LSHH set points
4. Inspect the dump valve operation

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: Sep 30 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

W/O crew discovered gathering line leak and notified Field Foreman. Line was isolated by Lease Operator.

Approximately 498 kcf released over a 14 hour period. This calculated to an above Tier 1 outdoor threshold. No effects to the public or environment.

WHAT WENT WRONG?:

Preventative Maintenance

Internal Corrosion

CORRECTIVE ACTIONS & RECOMMENDATIONS: n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

TIER 1 PROCESS SAFETY EVENTS

DATE: Oct 1 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

LO arrived at location and found standing oil inside the impermeable containment. LO walked the site and found the 2" load line on the back of the tank broke in two on the tank side of the ball valve.

WHAT WENT WRONG?:

Preventative Maintenance

Internal Corrosion

CORRECTIVE ACTIONS & RECOMMENDATIONS: n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: Oct 13 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Non-Agency Reportable Spill, Barnett, Wiley Pad – 284 bbls PW spilled/recovered and 16 bbls PO spilled/recovered from impermeable containment. Cause was internal corrosion on the back collar of the production tank.

WHAT WENT WRONG?:

Preventive Maintenance

Internal Corrosion

CORRECTIVE ACTIONS & RECOMMENDATIONS: n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

TIER 1 PROCESS SAFETY EVENTS

DATE: Nov 2 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Platform/Well Pad Flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Process Gas Release near the Battery on a 8" fiberglass gas gathering line.

WHAT WENT WRONG?:

Construction/Installation Less than Adequate.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Repair Plan was developed, approved by engineering and repairs were executed and completed in accordance with the repair plan.
- Pipeline flow could not be isolated in this case to allow for rapid response. Temporary flares had to be installed to divert gas flow from pipeline. Plan and activities are being worked to allow more rapid response time if this type of incident were to occur again in this field.

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

DATE: Nov 22 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Plant operators where inside the control room, heard a loud noise. The LEL and Low Suction alarms in the T-3 compressor building activated. The automated LEL sensors shut the plant down (T3 Compression & T3 Process). Compressor 3010, suction valve cap came off striking the turbo assembly.

WHAT WENT WRONG?:

Equipment / Material Failure

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS: n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

DATE: Dec 12 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The command center was contacted by local fire company about a gas release near the well pad. Lease operator responded and found a 1/2" nipple had broken on the glycol pump at the dehy unit.

WHAT WENT WRONG?:

- Equipment / Material Failure
- Preventative Maintenance

CORRECTIVE ACTIONS & RECOMMENDATIONS: n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Dec 23 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Night operator discovered ruptured FWKO inside containment that had released 65 bbls of fluid, 40 bbls (5 oil, 35 prod. water) inside containment and 25 bbls (5 oil, 20 prod. water) outside containment. All fluids were recovered and clean-up completed the day of event. Investigation ongoing.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

Equipment / Material Failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS: n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

DATE: Dec 23 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A pipeline failure occurred at a road crossing. Automated safety valves shut in the only two wells flowing into the gathering system. Gas released was estimated at 520 KCF and ignited at some point during the event.

WHAT WENT WRONG?:

Equipment / Material Failure

CORRECTIVE ACTIONS & RECOMMENDATIONS: n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: Jan 24 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Central Tank Battery. Heater treater fire tube burner had a leak and caused fire and spill of 0.81 barrels crude oil and 4.08 barrels produced water to land.

TIER 1 PROCESS SAFETY EVENTS

Volunteer Fire Department was called out to extinguish fire after gas supply to all wells were isolated. After approximately 45 minutes to an hour after the initial spraying down the heater, the fire department left location. No injuries occurred and the battery is secure.

Initial notification to NMOCD and State Land Office was immediately made.

LOC Severity Points: 1 pt.

WHAT WENT WRONG?:

Heater Treater fire tube burner had leak.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Unknown

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

DATE: Feb 2 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Unspecified

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Field Specialist detected a strong H₂S odour while driving his beat. While surveying the area, he noticed a gas cloud at the injection pad and isolated the line. He notified his supervisor who called the HES Field Specialist. The line was bled down while the area was monitored for H₂S from a distance. After the line was bled down gas-testing confirmed the area was clear and zero pressure was attained.

WHAT WENT WRONG?:

Unknown

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Unknown

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

NONE: None: Unspecified

TIER 1 PROCESS SAFETY EVENTS

DATE: Mar 6 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Platform/Well Pad Flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 10:28 am a call was received that there was a CO2 line leak near a production well. Upon further investigation, the leak was determined to be in the CO2 injection system. The field specialist closed the lateral isolation valve at the trunk line. Based on the gas plant discharge pressure trend, the leak was determined to begin at 10:10 am and finish at 11:51 am. Afterwards, the field specialists inspected the 4 wells on that trunk line lateral. 394AW and 392W were on H2O, so there was no action required. The 307W was on CO2 and was immediately shut in. The 444W was abandoned. The leak location occurred on a deadleg as the 444W was the last well on the lateral. It was calculated that 0.43 MMSCF gas was released to air.

LOC Severity Points: 3 pts

WHAT WENT WRONG?:

Leak

CORRECTIVE ACTIONS & RECOMMENDATIONS:

unknown

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: Mar 14 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Central Vacuum Unit gas line leak resulting in Tier 1 LOC with a total release volume of 711 MSCF.

Immediate notification to NMOCD was made and C128 was submitted.

LOC Severity Points: 1 pt.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

Unknown

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Unknown

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

DATE: Jun 26 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A flow line leak in an underground line occurred due to improperly torqued bolts. 2.1 MMSCF of gas was released to the air. Tier 1 LOC Spill/Release due to Flammable Gas released.

LOC Severity Points: 3 pts

WHAT WENT WRONG?:

Improperly torqued bolts.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Unknown.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

TIER 1 PROCESS SAFETY EVENTS

DATE: Nov 9 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Start-up

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Responded to callout to well pad. An LOC occurred after rupture of downstream piping following start up after a shutdown. MCV dump body and outlet of meter eroded through with remnants of sand and fluid inside GPU. Pad was already remotely ESD'd.

WHAT WENT WRONG?:

Ruptured pipe.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Unknown.

BARRIERS:

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

DATE: Sep 26 2020

COUNTRY: USA

FUNCTION: Drilling and Completion Operations

NUMBER DEATHS: 0

ACTIVITY: Drilling

MODE OF OPERATION: Other

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

A Contractor employee was working alone on the Residual gas Riser "Y" Strainer. The task was ordered due to a concern from Contractor and the Frac Site Leadership that there may be debris and liquid in the residual gas line. While the Frac was between stages, the Contractor employee closed the valve for the residue gas line and depressurized the residue side of the setup. Once the line was depressurized, the IP completed his task and reassembled the residue strainer. The IP then pressured up the system and returned to the strainer to gather his tools. The IP was in the process of bending over to collect his tools when the end cap gasket failed. The IP was struck on the right side of his face by an unknown object. During the investigation interview, the IP was not able to

TIER 1 PROCESS SAFETY EVENTS

definitively identify what struck his face. The IP was escorted from the scene of the incident to the HSE trailer by Frac Site employees. The IP was able to ambulate under his own power.

WHAT WENT WRONG?:

End cap gasket failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Unknown.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

DATE: Aug 13 2020

COUNTRY: USA

FUNCTION: Drilling and Completion Operations

NUMBER DEATHS: 0

ACTIVITY: Drilling

MODE OF OPERATION: Completion

POINT OF RELEASE: Equipment: Pump

CATEGORIES:

Fire/Explosion damage >\$100,000 direct cost to the company

INCIDENT DESCRIPTION:

During high pressure frac operations, the mechanical failure of a bearing in the power end of a pump caused the ring gear to break away from the crank shank. As a result, the force broke open the power end cover, exposing lube oil and sparking a fire. The fire spread across the adjacent equipment resulting in damage to eight frac pumps. The location personnel were accounted for and evacuated to a safe distance. Local emergency services were called and extinguished the fire safely.

WHAT WENT WRONG?:

- Bearing in power end of frac pump failed and caused ring gear to break away from crank shaft. As a result, the power cover separated and exposed lube oil which subsequently ignited.
- Pump operator was unfamiliar with proper operating parameters of lube oil temperature. Furthermore, contractor lacked an SOP of acceptable operating parameters for the lube oil temperature of the pump and instead relied on incorrectly applied knowledge of operating parameters of lube oil temperature.
- Equipment spacing allowed lube oil fire to spread and escalate to adjacent equipment, resulting in Tier 1 classification based on asset damage.
- High temperature alarm in pump operator's control area signalled bearing malfunction. However, pump operator failed to effectively acknowledge the alarm while operating the pump.

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Review asset's emergency response plan (ERP) with local authorities and ensure the expectation on potential scenario outcomes are realistic, especially regarding fire response and capability in your area. Ensure those detailed scenarios are documented and drilled within the asset's ERP.
- Ensure all personnel understand and are aligned with the alarm management process and response for all equipment and vendors on sites.
- Ensure sites utilize equipment-specific berms and reinforce the importance of using spacing guidelines during Management of Change (MOC) and work unit onboarding sessions.
- Evaluate quality, depth, and frequency of drills conducted on location.
- Review roles and responsibilities during emergency and shut in steps for all equipment during planned drills.
- Test depth of drills for realistic expectations of worst case scenarios that may be encountered during an emergency.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Ignition Control

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

Hardware Barrier Failures: Emergency Response Equipment and Systems

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Human Barrier Failures: Response to emergencies

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

TIER 1 PROCESS SAFETY EVENTS

NORTH AMERICA OFFSHORE

DATE: Jul 15 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Start-up

POINT OF RELEASE: Equipment: Compressor/blower/fan

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Gas release observed while starting up standby compressor 1. The compressor was started after a compression trip and inability of getting production 2 compressor started.

WHAT WENT WRONG?:

Level bridge with drain valves left open on the #1 Scrubber

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Operate in accordance to procedures.

Manage site and supervisors inspections .

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate communication

TIER 1 PROCESS SAFETY EVENTS

DATE: Nov 25 2020

COUNTRY: USA

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Heat exchanger

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Employee identified a gas leak. The employee used the field e-stop to shut down and secure the compressor, and appropriate blowdown. The leak was on an air cooled heat exchanger (fin fan) for the discharge of the compressor. Process Conditions at the time of release were 1250psig and 305F in Natural Gas service. No alarms were sounded. The wind was from the south, and the leak was at the northern end of the platform, so the released gas did not cross any gas detectors.

WHAT WENT WRONG?:

External corrosion caused the wall thickness of the cooler tubes on the inlet (hot) end, bottom row, to be significantly reduced. Once the coating debonded from the tubes due to high temperatures, the external corrosion was accelerated because water condensed on them due to the intermittent use of the cooler.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

No learning shared at this stage

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate communication

TIER 1 PROCESS SAFETY EVENTS

RUSSIA & CENTRAL ASIA ONSHORE

DATE: Jun 20 2020

COUNTRY: Azerbaijan

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Temporary

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

During excavation works, there was damage to a pipeline and a gas release occurred.

WHAT WENT WRONG?:

Despite the pipeline being closed to start maintenance, there was still extra gas in it which resulted in a fire.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Double check energy isolation, provide training/information/instructions to workforce, check exactly if the pipeline is fully empty.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: Aug 7 2020

COUNTRY: Azerbaijan

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

Ignition of flammable liquid during maintenance works on pipeline leading to tank.

WHAT WENT WRONG?:

Flammable liquid was not unloaded fully from the tank.

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Failure of energy isolation instructions/training/risk appraisal/lack of knowledge about unit working with.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Ignition Control

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: Nov 24 2020

COUNTRY: Kazakhstan

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Over-flowed oil was observed from a pit.

Oil was coming from a leaking line connecting well.

A Hot Oil Treatment of the flowline was just been completed.

WHAT WENT WRONG?:

Mechanical failure of cross-over at flange connection inside the pit.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

TIER 1 PROCESS SAFETY EVENTS

RUSSIA & CENTRAL ASIA OFFSHORE

No narrative reports were provided

SOUTH & CENTRAL AMERICA ONSHORE

DATE: Apr 23 2020

COUNTRY: Argentina

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Planned shutdown

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint

INCIDENT DESCRIPTION:

Flowline rupture while closing and depressurization of the well in order to repair a gas leak.

WHAT WENT WRONG?:

Operational Discipline.

Asset Integrity issues.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Do not start any work over the well until you have confirmed all the actuated valves are in the right position according to the actual procedure.

BARRIERS: No Barriers Allocated

CAUSAL FACTORS: No Causal Factors Allocated

DATE: Jan 29 2020

COUNTRY: Argentina

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

While loading a truck with capacity of 35 cubic metres, the system was programmed to load 37 cubic metres, resulting in an overfill of the truck. Operator shut loading operation. Clean-up initiated immediately. No environmental impact.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

- Training less than adequate
- Task Execution/Procedure Not followed for Other Reasons

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Review of the procedure with the operators of the area. Look for technology to reduce human dependency
- Talk to contractor to limit trucks entering with same capacity
- Fixed positions for experienced people only

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

DATE: Aug 7 2020

COUNTRY: Argentina

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Loading bay: While loading oil to a tank truck, the driver observed that the oil was overflowing. The driver approached the emergency button but he could not access it because the oil was splashing in the area. Another driver in the area notified the operator who initiated emergency shutdown of the loading area.

WHAT WENT WRONG?:

- Modification of tank without proper engineering and change management
- Procedure not followed by driver to open all caps on top

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Verify that every new trailer at the operation is fulfilling the requirements of certification and PM program
- Install an automatic overfill Shut Down system
- Relocate the emergency stop button and the View Panel of SCADA to an accessible sector.
- Review and update procedure for oil loading.
- Upgrade Loading Bay's Secondary Containment
- Train relevant personnel

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

TIER 1 PROCESS SAFETY EVENTS

DATE: Oct 15 2020

COUNTRY: Argentina

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Platform/Well Pad Flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Contractor personnel observed that water was sprouting from the ground between the plant and the well pad; immediately heads to the control room and report the plant operator. The operator proceeded to close the wells. Remediation started immediately. Nobody was hurt.

WHAT WENT WRONG?: n/a

CORRECTIVE ACTIONS & RECOMMENDATIONS: n/a

BARRIERS: No Barriers Allocated

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: Sep 30 2020

COUNTRY: Argentina

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

CATEGORIES:

PRD release to atmosphere above threshold in any 1 hour period and results in:

- liquid carryover or
- discharge to a potentially unsafe location or
- onsite shelter in place or
- public protective measure (e.g. road closure)

INCIDENT DESCRIPTION:

By performing compressor rotation (K-100-F), by alleviating towards Flare and encountering KOD Level (KO-700) spraying (SPRY) occurred in Flare + Liquid Burn.

WHAT WENT WRONG?:

Spill / Emission / Discharge to the Uncontrolled Environment

TIER 1 PROCESS SAFETY EVENTS

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Generate an Operational Instructions for Monitoring Levels in Containers Before / During Start-up or Field Manoeuvres.

Generate an MDC that contemplates the evaluation of the equipment associated with the Venting System (KOD-FLARE) in order to include criteria and / or level measurement standards that allow to add a philosophy / control routine for general SHUT DOWN plant.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

DATE: May 6 2020

COUNTRY: Argentina

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Upset

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)

CATEGORIES:

PRD release to atmosphere above threshold in any 1 hour period and results in:

- liquid carryover or
- discharge to a potentially unsafe location or
- onsite shelter in place or
- public protective measure (e.g. road closure)

INCIDENT DESCRIPTION:

Ruptured disc caused loss of containment in an early production facility.

WHAT WENT WRONG?:

Maintenance

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

TIER 1 PROCESS SAFETY EVENTS

BARRIERS:

Hardware Barrier Failures: Detection Systems

Hardware Barrier Failures: Protection Systems - including deluge and fire water systems

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Policies, standards and objectives

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

DATE: May 19 2020

COUNTRY: Argentina

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Pipeline breakage spill 10" Inter Plants (Cutting Plant 23 T – PTC PT), in plant headboard in heater enclosure.

WHAT WENT WRONG?:

Pipeline integrity

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

TIER 1 PROCESS SAFETY EVENTS

DATE: Mar 8 2020

COUNTRY: Argentina

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At about 10.00hrs, during pumping, there was a loss of containment due to the rupture of a 3" line upstream of the metering gauge.

WHAT WENT WRONG?:

Line breakage.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Jan 14 2020

COUNTRY: Argentina

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 06:00 am, during the operator's routine round, he noticed the overflow of the oil storage tank TK-101, at the oilfield oil separation facility. The oil was contained in its secondary containment barrier (dike).

The operators were supervising the normal operation of the facility. Water was being pumped from the gun barrel tank (TK-105), which is connected to the oil storage tank (TK-101), when the main water pump failed. TK-105 level started to rise until it started to send its fluids to TK-101 until its overflow.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

- The main water pump (gun barrel tank) failed. It did not have a maintenance routine.
- There are no high level alarms for operators visualizations. Poor instrumentation and control system and design in the facility (automatization).
- There are no specific instructions for operators routine rounds in order to detect deviations, nor operator procedures. Operators weren't properly trained.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Define and implement maintenance routine for the pumps and any other equipment (if necessary).
- Develop the SOP for the facility and checklists for the routine rounds.
- Develop and implement a training program for the operators (SOP).

Nowadays the facility has been shut down and it is no longer operative (marginal facility). However, if it becomes operative in the future, recommendations in order to improve the facility's instrumentation & control system will be implemented.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Dec 19 2020

COUNTRY: Brazil

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

LOPC of oily emulsion (BSW 50%) by line drain valve in onshore production well.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

Valves open incorrectly and absence of plug in drain of PIG launcher. Failure to pass service and communication.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Perform training recycling, review standards.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Policies, standards and objectives

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: Aug 12 2020

COUNTRY: Brazil

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period.

INCIDENT DESCRIPTION:

LOPC of gas through a valve in production line of a well.

WHAT WENT WRONG?:

The release occurred due to corrosion in valve body

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Improve maintenance plan for valves in remote wells.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

TIER 1 PROCESS SAFETY EVENTS

DATE: Feb 8 2020

COUNTRY: Brazil

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

After completing loading a water tanker, the operator left a valve open between the storage tanks and the separator, allowing flow from the tanks to the OW separator and resulting in an overflow of the separator.

WHAT WENT WRONG?:

- Failure to follow written work instructions, which were in any case not specific enough.
- Effects of a new Water Treatment Plant not assessed for impacts on Operator task execution and workload.
- Previous incidents not investigated thoroughly and lessons not learned.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Ensure that critical tasks are identified.
- Work instructions are in place known and followed by relevant personnel.
- Supervisors verify they are being followed.
- Management oversight is in place.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Violation unintentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

TIER 1 PROCESS SAFETY EVENTS

DATE: Feb 21 2020

COUNTRY: Brazil

FUNCTION: Drilling and Completion Operations

NUMBER DEATHS: 0

ACTIVITY: Drilling

MODE OF OPERATION: Completion

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Flexible hose/piping

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

During the removal of the drilling column from well, the driller interrupted the manoeuvre to completion well. To perform this operation, centrifugal pumps aligned directly to well through a 2" hose were used, but without success. The team involved attributed the lack of circulation to air intake at pumps. Then, the mud pump was aligned and the first drive was carried out. In the 2nd attempt to purge air from line with a mud pump, there was an abrupt uplift of the end of hose and it whipped, launching two roughnecks onto the floor of the platform. One of the roughnecks was struck on the upper and lower limbs, causing fractures and physical damage.

WHAT WENT WRONG?:

Operation performed in disagreement with standards

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Conducting training refresher on the drilling operation team regarding operational standards

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

TIER 1 PROCESS SAFETY EVENTS

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: Aug 15 2020

COUNTRY: Brazil

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During cargo offload operation Cargo Operator on duty watch reported a leak on discharge line of electric pump at a pipework joint weld seam. Cargo offload operations immediately stopped and header drained back to the cargo tanks and system isolated/depressurized.

Investigation commenced to determine the failure mechanism at the pipework weld seam leak location and condition of associated pipework.

WHAT WENT WRONG?:

Corrosion of carbon steel pipe led to pipe leak

CORRECTIVE ACTIONS & RECOMMENDATIONS:

No learning lessons available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

DATE: Oct 16 2020

COUNTRY: Brazil

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

There was a leak during the removal of tapping equipment in a hot water line and 2 employees were hit and burned.

TIER 1 PROCESS SAFETY EVENTS

WHAT WENT WRONG?:

Failure to comply with procedures and standards, failure to use adequate protective equipment, lack of training and supervision during operation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review contractual clauses regarding training of employees hired for tapping service on off-shore platforms, improve supervision of services for job permission and to inspect use of appropriate protective equipment.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Nov 8 2020

COUNTRY: Brazil

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Rupture of tubing in instrument socket in suction of the 2nd stage of the export compressor.

WHAT WENT WRONG?:

Excessive line vibration due to construction and assembly failure. The arrangement is long and without adequate support, contrary to the recommended by the manufacturer's catalogue.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Perform a vibration study to check for possible contribution to new occurrences on similar lines and provide adjustments, if necessary.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Detection Systems

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

TIER 1 PROCESS SAFETY EVENTS

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Jun 22 2020

COUNTRY: Brazil

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Loss of crude oil containment occurred through tubing of oil separator vent.

WHAT WENT WRONG?:

Failure to install tubing and no maintenance plan of equipment.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Check maintenance plans for tubing connected to vents of pressure vessels, adjust installation of tubing according to information from the manufacturers.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Apr 23 2020

COUNTRY: Brazil

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

LOPC of natural gas by PV Breaker system from cargo tanks.

WHAT WENT WRONG?:

Intense pressure transient in the system and reduction of water level in PV Breaker.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Importance of operational employees knowing in depth operation and functionality of equipment under their responsibility. This becomes even more relevant when equipment (or system) is classified as a “critical operational safety system”.

BARRIERS:

Hardware Barrier Failures: Detection Systems

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

Management System Element Barrier Failure: Policies, standards and objectives

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

TIER 1 PROCESS SAFETY EVENTS

DATE: Jan 21 2020

COUNTRY: Brazil

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Subsea: Subsea pipeline/flowline

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Breaking of the gas injection line.

WHAT WENT WRONG?:

Stress corrosion cracking by CO₂ and inspection techniques employed inefficient to ensure the detection of tensile armour degradation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review inspection standard for flexible ducts susceptible to SCC CO₂ with a focus on torsion detection; ensure that leak tightness tests are carried out in all stages prior to the operation.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Policies, standards and objectives

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

TIER 1 PROCESS SAFETY EVENTS

DATE: Dec 24 2020

COUNTRY: Brazil

FUNCTION: Production

NUMBER DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Unspecified

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

INCIDENT DESCRIPTION:

Detection of gas leakage in a closed drain line on the production deck.

WHAT WENT WRONG?:

Operating standard of closed drainage system allows connection of different pressure class systems without adequate mitigation of risk of blockade followed by overpressure and change management not performed to operate the equipment without safety devices.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review the Closed Drain Storage standard, to make clear the need to withdraw from operation and prior depressurization of equipment, before draining for closed drain, as determined by the Design Philosophy, conduct refresher training on the change management procedure and improve definition of criteria for indication of a technical team for elaboration and review of operational standards based on criteria such as system or criticality.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Protection Systems - including deluge and fire water systems

Hardware Barrier Failures: Emergency Response Equipment and Systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

TIER 1 PROCESS SAFETY EVENTS

DATE: Jan 16 2020

COUNTRY: Brazil

FUNCTION: Drilling and Completion Operations

NUMBER DEATHS: 0

ACTIVITY: Drilling

MODE OF OPERATION: Drilling

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A leakage of 2.38 cubic metres (15 bbl) of diesel was observed in a high pressure pump relief valve flange connection. The leak was retained in the containment area of the tank room.

WHAT WENT WRONG?:

- Failure in Planning: The activity of pumping fluid from cleaning the well was carried out without proper planning. The valve manufacturer's manual was not consulted to check the operational limitations of the equipment.
- Management of change (MOC) Failure: The risk analysis of the well did not foresee the performance of the activity. Change management was not carried out for the evaluation of unforeseen activity.
- Risk Analysis Failure: No specific risk analysis was performed for the Well Test plant's diesel pumping operation for cleaning the sand from the well in a closed circuit, so the possibility of solid particles causing the seal to break was not identified .

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Establish criteria and standardize systematic service requests through the hierarchical line regarding the planning and execution of non-routine activities that may generate new risks in the installation.
2. Hold a specific non-routine meeting to define the criteria and control measures for pumping flammable fluids.
3. Reinforce in the pre-shipment meetings the importance of carrying out change management when necessary.
4. Reinforce for the board leadership the requirements of the Change Management system.
5. Carry out risk analysis of the non-routine activity of circulation of return fluids from the well, considering the debris and particles contained in the fluid that may cause damage to the equipment.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

FATAL INCIDENT REPORTS CLASSIFIED AS TIER 1 PROCESS SAFETY EVENTS – 2020

IOGP has been gathering Fatal incident descriptions from its Members since 1991. These include upstream fatal incidents reported both by companies and contractors and, more recently, incidents involving 3rd party fatalities. Incident reports are categorised by region, country, location (onshore/offshore), cause and activity at the time of the incident, and Life-Saving Rule. For each incident IP details of function, employer (company or contractor), age and occupation are collected.

The database of fatal incidents categorised as Tier 1 PSE, or as PSE related, is now available and searchable at <https://data.iogp.org/ProcessSafety/FatalIncidents>.

For the entire database of fatal incidents go to <https://data.iogp.org/Safety/FatalIncidents>.

This database is a tool for learning and should not be considered a complete record of fatal incidents in the upstream oil industry or the IOGP Membership.

FATAL INCIDENT REPORTS CLASSIFIED AS TIER 1 PROCESS SAFETY EVENTS – 2020

AFRICA ONSHORE

DATE: Dec 13 2020

COUNTRY: Nigeria

FUNCTION: Production

NUMBER OF DEATHS: 1

CAUSE: Explosion, **ACTIVITY:** Production operations

RULE: Unspecified

BODY PART: Unspecified

NATURE OF INJURY: Burn

Employer: Company

Occupation: Other

NARRATIVE:

Information not available

WHAT WENT WRONG:

Information not available

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Information not available

CAUSAL FACTORS: No Causal Factors Allocated

AFRICA OFFSHORE

No fatal incidents classified as Tier 1 PSE reported

ASIA / AUSTRALASIA ONSHORE / OFFSHORE

No fatal incidents classified as Tier 1 PSE reported

EUROPE ONSHORE / OFFSHORE

No fatal incidents classified as Tier 1 PSE reported

FATAL INCIDENT REPORTS CLASSIFIED AS TIER 1 PROCESS SAFETY EVENTS – 2020

MIDDLE EAST ONSHORE

DATE: Jun 11 2020

COUNTRY: Kuwait

FUNCTION: Drilling

NUMBER OF DEATHS: 2

CAUSE: Explosion, **ACTIVITY:** Drilling, workover, well services

RULE: Line of fire

BODY PART: Unspecified

NATURE OF INJURY: Burn

Employer: Contractor Occupation: Drilling/Well Servicing Operator

Employer: Contractor Occupation: Drilling/Well Servicing Operator

NARRATIVE:

- Rig Crew started gas bleeding at 02:15 hrs. from well casing to the Far Pit.
- After finishing gas bleeding, while preparing to kill the well, a flash fire occurred at the pit, which caused severely burning two roustabouts stationed near the pit. The fire was extinguished on casing valve closure.
- Both injured persons (IPs) were transferred to a hospital. One died on the date of incident, with the second individual dying several days later.

WHAT WENT WRONG:

- A sudden high velocity fluid surge from the well.
- Fire in the pit likely from ignition due to friction in the pit or probable static electric discharge between gas and the pit discharge line.
- Misperception of risk by crew.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- No crew is to be allowed near flare pit during bleed off operations.
- Ensure periodic inspection and maintenance of adequately designed grounding earthing for effective dissipation of static electricity accumulated on the vent/drain pipe up to the pit.
- Enhancement of awareness about static electricity hazards in flammable atmospheres.
- Assess and manage Covid-19 impact on monitoring as well as execution of critical activities.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST OFFSHORE

No fatal incidents classified as Tier 1 PSE reported

FATAL INCIDENT REPORTS CLASSIFIED AS TIER 1 PROCESS SAFETY EVENTS – 2020

NORTH AMERICA ONSHORE / OFFSHORE

No fatal incidents classified as Tier 1 PSE reported

RUSSIA & CENTRAL ASIA ONSHORE / OFFSHORE

No fatal incidents classified as Tier 1 PSE reported

SOUTH & CENTRAL AMERICA ONSHORE / OFFSHORE

No fatal incidents classified as Tier 1 PSE reported

FATAL INCIDENT REPORTS RELATED TO PROCESS SAFETY BUT NOT CLASSIFIED AS TIER 1 PROCESS SAFETY EVENTS 2020

None of the fatal incidents reported to IOGP in 2020 were categorized as process safety related.

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

High Potential Events are defined to be any incident or near miss that could, in other circumstances, have realistically resulted in one or more fatalities.

IOGP has been gathering high potential descriptions from its Members since 2000. These include upstream significant events reported both by companies and contractors. Event reports are categorised by region, country, location (onshore/offshore), cause and activity at the time of the event, and Life-Saving Rule.

The database of high potential events categorised as Tier 1 PSE, or as PSE related, is now available and searchable at <https://data.iogp.org/ProcessSafety/HighPotentialEvents>.

For the entire database of high potential events go to <https://data.iogp.org/Safety/HighPotentialEvents>.

Note that high potential events requested are only those with the highest learning value, so the information shown does not represent all reportable events.

This database is a tool for learning and should not be considered a complete record of high potential events in the upstream oil industry or the IOGP Membership.

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

AFRICA ONSHORE

DATE: Jul 12 2020

COUNTRY: Tunisia

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Excavation, trenching, ground disturbance

RULE: Work authorization

NARRATIVE:

On Friday June 12, 2020, during sand removal activities for an escape route commonly understood as a “patrol road” the underground 6” gas flow line (FL) was punctured by a front loaders teeth as it was levelling the ground. At the time of the incident, the pressure of the gas 6” FL and the methanol 2” FL were respectively approximately 54 barg and approximately 285 barg respectively. A vapor cloud was immediately generated without ignition. Luckily, no unconfined Vapor Cloud Explosion (UVCE) or jet fire occurred. The involved contractor team comprising of five (5) persons evacuated the site and moved to a safe location. No injuries occurred.

WHAT WENT WRONG:

- Unidentified (unmarked) underground services in immediate proximity to sand removal operations.
- Absence of warning tape on top of buried services. The investigation revealed few non-conformances in the construction and gaps vs the as-built drawings
- Missing key marker pole and absence of warning tapes on top of the FL or Fiber Optic Cable (FOC).
- Widening the escape route to more than required allowed for repetitive access by vehicles over time and the perception that this was a “NORMAL” or “ROUTINE” road that can be subject to normal grading by heavy equipment

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure better Knowledge management from “EXECUTE” to “OPERATE” and prevent knowhow losses: Operation team should ensure more synergy and consultation with remaining project organization during “OPERATE” phase of the project mainly in future MoC or scope of work approval.
- The Management of construction change via technical queries (TQ's) by EPCC is deemed inefficient than a standard Company MoC with a supporting risk assessment. In this particular change of the trench configuration, Projects should have provided additional warning and signalization equipment to denote the underground services which are, in this case, exceptionally close to the natural ground level, and were at high risk of impact by ground disturbance either natural or manmade.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate communication

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

AFRICA OFFSHORE

DATE: Aug 5 2020

COUNTRY: Angola

FUNCTION: Production

CAUSE: Unspecified - Other

ACTIVITY: Maintenance, inspection, testing

RULE: Other issue

NARRATIVE:

During routine testing, a diesel engine driving a fire water pump caught on fire. The fire initiated from an LOPC created by two pin holes in the diesel supply line. The pin holes were the result of external corrosion.

WHAT WENT WRONG:

During building of the vessel, the diesel engine OEM issued a technical service bulletin calling for the replacement of all the diesel fuel lines on the engine. The bulletin required the replacement lines to be painted and this step was never completed. The unpainted lines resulted in an accelerated corrosion rate.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Verify that a process to manage vendor service bulletins and modifications is in place during the execute phase of major projects and confirm whether it defines how ownership is handed over to equipment owners within operations.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

ASIA / AUSTRALASIA ONSHORE

DATE: Jul 20 2020

COUNTRY: Australia

FUNCTION: Production

CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature

ACTIVITY: Maintenance, inspection, testing

RULE: Unspecified

NARRATIVE:

An operator attended a well site on routine rounds. On approach, the operator noticed the HPU (Hydraulic Power Unit) making abnormal noises and began troubleshooting the source of the noise, during which operator's Personal Gas Monitor (PGM) alarmed on high Carbon Monoxide (CO). The operator continued troubleshooting to identify and document the source of the leak as the engine exhaust flange. On return to his vehicle, the operator felt increasingly unwell, and initiated emergency response. The operator was later hospitalized for exhaust inhalation.

WHAT WENT WRONG:

The engine exhaust flange was loose (chattering) at the 'donut' gasket (spring loaded flange) between the engine manifold and exhaust pipework, causing a release of exhaust gases. The operator continued to troubleshoot the HPU after acknowledging a PGM CO high level alarm. The operator also felt safe to continue the task, where the risk of Carbon Monoxide exposure from the exhaust was perceived as low.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Lessons Learnt:

- Awareness risks posed by exposure to CO and exhaust gases.
- Do workers know PGM alarm limits are for CO?
- Does our relevant documentation (SOP/PET/HAZOP) adequately cover CO and exhaust hazards?
- How are we responding when our gas detectors go into alarm? Does our response change depending on what gas is present (CH₄, CO, O₂, H₂S)? Have we begun to normalize this risk in some areas of our operations?
- Do teams feel empowered to escalate concerns and activate their distress button in an emergency?

Recommendations:

- Ensure CO risk is recognized and understood across the business, including not only HPUs (combustion engines), but also for gas driven and temporary equipment,
- Determine and document flange management requirements for spring loaded joint types,
- Rollout investigation learnings on CO hazards and gas monitoring.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

ASIA / AUSTRALASIA OFFSHORE

DATE: May 11 2020

COUNTRY: Australia

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Production operations

RULE: Other issue

NARRATIVE:

Oil production FPSO reeled offtake hose suffered sudden catastrophic failure whilst stored on hose reel.

Significant crude oil LoPC and vapour cloud all contained within hose reel bund.

Hose segments were all 10 years old (stored outside) and in service for 5 or 10 years.

WHAT WENT WRONG:

External inspection was not targeted at close visual of flange joint stud bolt pockets where indication of water ingress was visible on several hose sections.

Water ingress caused internal steel armour wires to corrode which was undetected during annual general visual inspection.

Following the event, we learned other operators had experienced similar failures which could have informed our inspection scope.

Hose was 20" diameter (many are 16" diameter) on spool with minimum diameter of 7m (many are 8+m diameter), the tight bend radius may have contributed to the sudden internal wire failure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

No injuries or damage to the environment with Emergency Shut Down systems automatically triggered and the release captured in dedicated bund.

Maintenance and Inspection was up to date, however maintenance strategy was ineffective at detecting early signs of water ingress causing internal steel wires to corrode. Be cautious about component replacement times based on inspection results.

Make full use of external learnings from manufacturer and other users in updating Maintenance Strategies.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

DATE: May 6 2020

COUNTRY: Malaysia

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Maintenance, inspection, testing

RULE: Line of fire

NARRATIVE:

IP suffered an injury to his inner thigh while removing a Dry Gas Seal (DGS) from the IP/MP A Gas Compressor, suspected has a pressure trapped and sudden movement of DGS hitting the IP.

WHAT WENT WRONG:

Procedures – Followed Incorrectly

Training – Understanding Needs Improvement

Late communication

Quality Control – No inspection – Hold Point Not Performed

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure knowledge and competency of isolation philosophy are adequate prior executing the work. If there's clarity needed, suspend the work and seek technical support.
- Before breaking containment – always conduct the final check for all possible trapped pressure that may increase the risk of the activity.
- Ensure two-way communication between with all parties from both operators and service providers.
- If there's a change in work scope, re-conduct the safety and integrity verification to ensure all risk have been mitigated.
- If there's known valve passing, additional attention should be given especially when breaking containment activity is planned around the area.
- Review work area and body position prior work execution to minimize the risk.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Overexertion or improper position/posture for task

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

EUROPE ONSHORE

DATE: Aug 31 2020

COUNTRY: Romania

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Production operations

RULE: Other issue

NARRATIVE:

On 31.08.2020 around 16:50 the operator from Dehydration Station noticed the decrease of gas flow in the station. The 8 5/8" gas pipeline was sectioned by an excavator of a third party while executing works to install some communication cables.

WHAT WENT WRONG:

- a) Direct causes (substandard acts and conditions) – Gas pipeline damaged by a third party
- b) Underlying causes (contributing factors / pre-conditions)
 - Third party company performed unauthorized work around gas pipeline
 - Third party company did not requested authorization from authorities
 - Third party company considered no authorization needed for such work
 - Third party took the risk of unauthorized work based on experience of such work performed in the past
 - Third party company was not aware about pipeline markings - the Company was not aware about work performed around gas pipeline
 - Third party company did not requested approval from the Company
- c) Root causes
 - Third party did not comply with legal requirements regarding performing works around oil and gas facilities

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Reporting of any activity performed by third-party individuals and companies in the vicinity of the company's facilities and pipelines is important to prevent similar incidents. Increase personnel awareness and update the training on emergency shut down of oil and gas production facilities and pipelines

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate communication

EUROPE OFFSHORE

No high potential events classified as PSE

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

MIDDLE EAST ONSHORE

DATE: Oct 16 2020

COUNTRY: Oman

FUNCTION: Production

CAUSE: Unspecified - Other

ACTIVITY: Production operations

RULE: Work authorization

NARRATIVE:

ESP Well Started on 15th at 05:30 PM and line up to production while plant manifold battery limit valve was isolated & CSC (car-seal closed). Flow line pressure build up and over pressurized the line to maximum designed operating class #300(720 psi), 4 flanges at well test spool and plant manifold 6" isolation valve leaked. The leak noticed next day at 7:55 AM & found wellhead SSV not lined up.

WHAT WENT WRONG:

The immediate cause of failure which resulted in loss of containment, was overpressure of 6" flowline due to 6" ball valve at station manifold.

The two primary strategies for protecting against this overpressure hazard are wellhead ESD and downhole ESP (trip on high motor temperature). The Wellhead ESD valve was not lined up to the flowline, it was isolated by utilizing a cap and pump intake pressure did not reach trip set point.

Inadequate communication between field and plant operators.

Inadequate training and understanding of risk. Inadequate communication of SOPs.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Well start up SOP and check list will be reviewed.

Conduct line walk down.

Follow LOTO SOP.

Issue Standing Instruction (SI) to Production Operations to ensure that line walk down completed and all valves are lined up to station, so that overpressure condition cannot occur.

Conduct awareness sessions for all field and plant operators, maintenance on Human Factors aspects associated with Emergency Shutdown Valve operation as safety critical equipment.

Conduct an inspection of the Wellhead ESD systems

Conduct a Process Hazard Analysis (PHA) of ESDV systems to review design aspects associated with flowline overpressure due to valve closer/ block.

Review the installation and configuration of wellhead ESDV with Operation and Maintenance team.

Review training requirements for recently recruited assistant operators.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

MIDDLE EAST OFFSHORE

No high potential events classified as PSE were reported

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

NORTH AMERICA ONSHORE

DATE: Sep 29 2020

COUNTRY: Canada

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Production operations

RULE: Unspecified

NARRATIVE:

Fire Tube Failure resulting in a Fire. Liquid carry over on the flare stack resulting in ground fire

WHAT WENT WRONG:

Root cause(s) Maintenance management

- Inspection plan does not exist or is not implemented Root cause(s) Maintenance management - Maintenance plan does not exist or is not implemented Root cause(s) Procedures and control of operations
- Lack of procedure

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Generate FWKO & Treater Firetube Fire Emergency response instructions for the 12-19 Battery prior to Battery start up.

Generate Emergency response instructions for all other batteries in the asset for responding to abnormal (dark/black) smoke or fire from burner stacks in fire tube vessels. Conduct a field level workshop to determine if there are any additional scenarios where a shutdown procedure is required.

Develop the procedure(s) required based on the scenarios found.

Update maintenance PM (beyond current function testing) scope and steps of ESDVs to be in accordance with the requirements of the ESD Systems Performance Standard and manufacturer recommendations (whichever is more stringent).

Update Performance Standard as required.

Update current Firetube Repair Procedure.

Add firetubes to the Pressure Vessel Performance Standard (PC007). Include design and maintenance requirements.

Ensure firetubes are captured in both the CMMS and Integrity D

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

DATE: Jan 16 2020

COUNTRY: Canada

FUNCTION: Production

CAUSE: Unspecified - Other

ACTIVITY: Maintenance, inspection, testing

RULE: Other issue

NARRATIVE:

An explosion and subsequent flash fire occurred around two fuel gas letdown buildings due to an increase of the Lower Explosive Limits up to 20% inside of the buildings.

WHAT WENT WRONG:

1. Equipment Difficulty: Design specifications for nozzles and hi level indication needs improvement.
 - Issues with nozzle size for level transmitter was not sent to engineering, vessel also not equipped with any level alarm or shutdown.
2. Equipment Difficulty: Alarm Specifications Need Improvement:
 - LEL alarms are set at a low of 20% which led to operators being unaware of the levels.
3. Management Systems: No procedure:
 - No operating procedure in place to deal with high liquid levels in the fuel gas system.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Changes to be made to DCS to divert compressors into re-injection mode as result of any upset in refrigeration system.
2. Training to all operators on changes to system.
3. Changes to plant fire & gas system to alarm at 10% LEL and high level at 20%.
4. Add external audible and visual LEL alarms.
5. Car seal close manual bypass valve to prevent gas blow by.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

DATE: Apr 4 2020

COUNTRY: USA

FUNCTION: Production

CAUSE: Explosion

ACTIVITY: Production operations

RULE: Hot work

NARRATIVE:

While nearing mechanical completion on a (Saltwater Disposal) SWD project, contractor fiberglass crew was replacing a defective fitting on a line to a produced water tank. Following the manufacturer's procedure, the fiberglass crew used an open flame torch to heat and dry the fiberglass. As the open flame was applied to the piping, hydrocarbon vapours from the produced water in the tank ignited resulting in a deflagration inside the tank. The tank roof with valves, instrumentation and tank piping were separated from the tank wall and landed on the ground. Personnel evacuated and mustered without injury.

WHAT WENT WRONG:

1. Lack of Hazard Recognition – The introduction of Produced Water to the construction site (greenfield/ brownfield) was not clearly recognized as a new hazard.
2. Inadequate Communication Between Supervision and Crew – The introduction of PW was not mentioned during the repair crew's site orientation, nor did the crew mention using an open flame when explaining their job tasks.
3. Review Not Performed or Less Than Adequate JSA - Site JSA makes no mention of PW and the crew did not complete a task specific JSA.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Install signage and flagging indicating the introduction of hydrocarbons
- Update contractor orientation training to recognize hazards associated with hydrocarbons
- Provide refresher training on effectively communicating site transition from greenfield to brownfield
- Complete safety stand-down prior to introducing hydrocarbons at sites that are not at mechanical completion
- Update Hot Work safety information to include produced water as a potential hydrocarbon source

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

DATE: Jan 15 2020

COUNTRY: USA

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Production operations

RULE: Line of fire

NARRATIVE:

- On January 15th at 22:35, a trouble alarm came in on 'C' Recycle compressor; followed by failure of the recycle compressor resulting in a CO₂ release inside the recycle compressor building
- All five building CO₂ detectors maxed out at 20,000 ppm

WHAT WENT WRONG:

- Lab concluded studs failed through a combination of fatigue and overload failure as applied stress increased on un-cracked studs
- Most likely cause of bolt fatigue was misalignment of cylinder or inadequate torque of studs
- Existing compressor monitoring/alarm system not capable of providing early warning of failure

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Validate fit-for-service status of cylinder to distance piece studs and supports
- Retain controlled access to the building under SCBA per SBRA until fit-for-service validation is complete on the four identical recycle machines currently in operation
- Update processes and procedures for cylinder alignment, surveillance and maintenance
- Long term - review cylinder support design and evaluate monitoring/control system upgrades

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate maintenance/inspection/testing

NORTH AMERICA OFFSHORE

No high potential events classified as PSE were reported

RUSSIA & CENTRAL ASIA ONSHORE / OFFSHORE

No high potential events classified as PSE were reported

HIGH POTENTIAL EVENTS CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

SOUTH & CENTRAL AMERICA ONSHORE

DATE: May 12 2020

COUNTRY: Ecuador

FUNCTION: Production

CAUSE: Unspecified - Other

ACTIVITY: Production operations

RULE: Other issue

NARRATIVE:

During the GENERATOR B shutdown process, a leak is identified in the fuel inlet pipe (crude).

The pipe is disassembled and sent to the CPF for repair. The welder at CPF repairs the crack with electric welding and sent it back to the work site. The tubing is installed in GENERATOR B and plucked. Approximately 4 hours after its operation, another leak occurs in the same pipeline, so the Lead Operator decides to turn off GENERATOR B.

During the shutdown process of GENERATOR B, a shutdown occurs in the same GENERATOR.

WHAT WENT WRONG:

Inadequate Administration/Management of Change

Welding repair process out of the manufacturer's specifications there was not records of previous traceability of that repair. Not apply Management of Change Process.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

When apply any repairing process, verify all technical specifications by the manufacturer, follow up manuals and/or in the applicable procedures established by the Company. Make sure that all people involved know the instructions and procedures that must be implemented. Follow management of change procedure of the Company.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials & Products: Inadequate design/specification/management of change

SOUTH & CENTRAL AMERICA OFFSHORE

No high potential events classified as PSE were reported

HIGH POTENTIAL EVENTS RELATED TO PROCESS SAFETY BUT NOT CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

High Potential Events are defined to be any incident or near miss that could, in other circumstances, have realistically resulted in one or more fatalities.

IOGP has been gathering high potential descriptions from its Members since 2000. These include upstream significant events reported both by companies and contractors. Event reports are categorised by region, country, location (onshore/offshore), cause and activity at the time of the event, and Life-Saving Rule.

The database of high potential events categorised as Tier 1 PSE, or as PSE related, is now available and searchable at <https://data.iogp.org/ProcessSafety/HighPotentialEvents>.

For the entire database of high potential events go to <https://data.iogp.org/Safety/HighPotentialEvents>.

Note that high potential events requested are only those with the highest learning value, so the information shown does not represent all reportable events.

This database is a tool for learning and should not be considered a complete record of high potential events in the upstream oil industry or the IOGP Membership.

HIGH POTENTIAL EVENTS RELATED TO PROCESS SAFETY BUT NOT CLASSIFIED AS PROCESS SAFETY EVENTS – 2020

AFRICA ONSHORE / OFFSHORE

No high potential events classified as process safety related were reported

ASIA / AUSTRALASIA ONSHORE / OFFSHORE

No high potential events classified as process safety related were reported

EUROPE ONSHORE

DATE: Jun 9 2021

COUNTRY: Croatia

FUNCTION: Production

CAUSE: Unspecified - Other

ACTIVITY: Production operations

RULE: Unspecified

NARRATIVE:

A third party reported an oil leakage. A site inspection confirmed that there was a leakage on pipeline between 2 gathering stations. The recultivation and repair works were started immediately.

WHAT WENT WRONG:

Equipment reliability.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Improvement of inspection and pipeline maintenance.

CAUSAL FACTORS: No Causal Factors Allocated

EUROPE OFFSHORE

No high potential events classified as process safety related were reported

MIDDLE EAST ONSHORE / OFFSHORE

No high potential events classified as process safety related were reported

HIGH POTENTIAL EVENTS RELATED TO PROCESS SAFETY BUT NOT CLASSIFIED
AS PROCESS SAFETY EVENTS – 2020

NORTH AMERICA ONSHORE / OFFSHORE

No high potential events classified as process safety related were reported

RUSSIA & CENTRAL ASIA ONSHORE / OFFSHORE

No high potential events classified as process safety related were reported

SOUTH & CENTRAL AMERICA ONSHORE

DATE: Dec 16 2020

COUNTRY: Brazil

FUNCTION: Production

CAUSE: Unspecified

ACTIVITY: Drilling, workover, well services

RULE: Unspecified

NARRATIVE:

LOPC of oily emulsion due to a hole in the lower generatrix of the pipe that connects the cold wash tanks to the hot wash tanks.

WHAT WENT WRONG:

Under investigation.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Under investigation.

CAUSAL FACTORS: No Causal Factors Allocated

SOUTH & CENTRAL AMERICA OFFSHORE

No high potential events classified as process safety related were reported

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Fatal incidents and high
potential events that were also
PSE, and fatal incidents and high
potential events that were PSE-
related – 2020